CASEbot: A Conversational Agent for Structuring and Personalizing the Design of Self-Experiments in Personal Health

Sabrina Zaman Ishita*
Manning College of Information & Computer Sciences
University of Massachusetts Amherst
Amherst, Massachusetts, USA
sishita@umass.edu

Sidharth Kaliappan*
University of Massachusetts Amherst
Amherst, Massachusetts, USA
skaliappan@umass.edu

Mashrur Rashik
University of Massachusetts Amherst
Amherst, Massachusetts, USA
mrashik@cs.umass.edu

Daniel A. Epstein
Informatics
University of California, Irvine
Irvine, California, USA
epstein@ics.uci.edu

Ravi Karkar
Manning College of Information & Computer Sciences
University of Massachusetts Amherst
Amherst, Massachusetts, USA
rkarkar@cs.umass.edu

Abstract

Self-experimentation, or using tracked data to systematically answer health and wellbeing questions via hypothesis testing, has significant potential to support personal health. However, technological support for self-experimentation has focused on expert-designed self-experiments for specific health conditions, limiting people's ability to design their own rigorous experiments. To address this gap, we developed **CASE**bot (**C**onversation **A**gent for **S**elf-**E**xperimentation), an LLM-powered chatbot using a theory-driven approach to guide users through designing well-structured, personalized, and safe self-experiments. We conducted a within-subjects, mixed-methods study with 42 participants comparing **CASE**bot to a traditional worksheet-based approach. When formally comparing the experiment rigor and specificity, most participants designed better experiments using **CASE**bot. They appreciated **CASE**bot's conversational approach, which prompted them to surface everyday constraints and proactively raised safety concerns, but some found the platform too rigid in its recommendations. We discuss opportunities for future generative AI self-experimentation systems for health to balance structured guidance with user autonomy.

CCS Concepts

• **Human-centered computing** → *Interactive systems and tools*; **User studies**; **Empirical studies in HCI**.

Keywords

self-experimentation, self-tracking, personal informatics, conversational agent, health, wellbeing, n-of-1

*These authors contributed equally to this work.

This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI '26, Barcelona, Spain
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2278-3/26/04
https://doi.org/10.1145/3772318.3791551

ACM Reference Format:
Sabrina Zaman Ishita, Sidharth Kaliappan, Mashrur Rashik, Daniel A. Epstein, and Ravi Karkar. 2026. CASEbot: A Conversational Agent for Structuring and Personalizing the Design of Self-Experiments in Personal Health. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 26 pages. https://doi.org/10.1145/3772318.3791551

1 Introduction

Self-experimentation is the process of using one's own data to formally evaluate and answer questions about oneself (e.g., *Does drinking coffee after 2pm affect my sleep quality?*) [63, 72, 78]. Within HCI, research has often examined how technology, especially self-tracking technology, can assist with self-experimentation, such as by scaffolding experiment design and formalizing analysis in domains like sleep [23–25], irritable bowel syndrome [47], and behavior change [58]. However, research has regularly suggested that even people knowledgeable about data analysis encounter significant challenges with experiment design, such as collecting the data which is not useful or lacking in scientific rigor in experiment design [16]. Existing platforms address these pitfalls by having experts design experiments for users to select from a pre-approved list. While rigorous, this approach is unable to support individuals creating self-experiments tailored to their own goals. In this research, we seek to address this limitation by developing a **platform that makes the design of self-experiments more accessible by scaffolding the required expertise while providing necessary structure, personalization, and safeguards.**

The practice of self-experimentation encompasses multiple distinct phases – design, execution, and analysis [48]. In this research we focus primarily on the experiment design stage, where people plan the overall process they would undergo to collect data, analyze it, and act on the findings. Supporting effective experiment design is critical because similar to other personal informatics contexts [60], barriers and mistakes cascade through the later stages of executing and analyzing an experiment. Prior research has shown that poorly designed experiments

may result in people collecting data which does not support answering their question, data which cannot be reliably analyzed, or more data than they can reliably keep up with collecting [16]. Resolving such a range of constraints around the choices that need to be made during design is too difficult for a traditional application [25, 47]. However, large language models (LLM) are well suited for such ill-defined problems given their ability to interpret ambiguous requirements and resolve constraints through natural conversation [7, 55, 71, 85, 86]. The recent development of reasoning models [94] has enabled such LLM-conversational agents to go beyond simple querying to reacting to user input without explicit instructions. They can be used to collect necessary personal information and context, and guide an individual through the design of a valid self-experiment. This enables them to better navigate the gaps between what steps are needed to create an ideal self-experiment and the personal constraints that the user brings forth.

To support structuring and personalizing self-experiments in health and wellbeing, we developed **CASE**bot (**C**onversation **A**gent for **S**elf-**E**xperimentation), an LLM-powered chatbot using a theory-driven approach to guide users through **designing** well-structured, personalized, and safe self-experiments. Our approach builds on HCI models and frameworks for supporting experimentation with tracking data, understanding failure points, and enabling personally meaningful explorations [16, 34, 60, 70]. We also lean on decades of research on designing self-experiments [12, 28, 48, 50, 57, 72]. First, **CASE**bot provides **structure** by breaking down the design process into four dimensions: defining a clear *Hypothesis*, choosing an *Independent Variable* (the change) and *Dependent Variable* (the outcome), selecting an *Experiment Type*, and creating a detailed *Experiment Schedule*. Second, it supports **personalization** by asking about the user's health goal, lifestyle, and other constraints, incorporating that information into the plan. Finally, it enhances **safety** by performing checks on intervention dosage and relevant health background to avoid risky experiments. **CASE**bot was designed to adopt a conversational approach similar to a coach or a guide and lead the individual to an experiment designed to support their needs instead of simply presenting fully-formed self-experiment designs.

To evaluate our system, we conducted a within-subjects, mixed-methods study with 42 participants who created experiments in physical activity, diet, and sleep, three commonly-studied experiment domains [22, 70, 83]. We compared the experimental plans created using **CASE**bot against those from a traditional structured worksheet which introduced experiment steps similarly to **CASE**bot without interactivity and explicit personalization. Participants found the worksheet to be a valuable tool for scaffolding the experiment design process, but found it lacking feedback on the rigor and safety of experiments. Conversely, they appreciated the support **CASE**bot provided towards breaking down the design process and personalizing the experiment design and schedule in light of their routines and constraints. Analyzing the chat transcripts, we found that **CASE**bot was able to raise concerns in instances where the intervention dosage may be harmful, thereby enhancing the safety of the experiments. Given the difficulty in comparing self-experiments, particularly when they are focused on a range of topics (e.g., testing the impact of consuming fruits on mood vs the effect of screen time on sleep quality); we leveraged common criteria for evaluating experiments, based on prior work, to develop a scoring rubric to enable quantitative comparison of experiments. Using this rubric, we found that nearly all participants [39 out of 42 (92.86%)] created better experiments with **CASE**bot when compared to worksheet with an average score improvement of 19.20%.

Overall, this paper makes three contributions:

- The design of a Generative Artificial Intelligence (genAI) enabled conversational platform, **CASE**bot, aimed at supporting novices through designing a wide array of self-experiments by integrating our principles of structure, personalization and safety.
- Findings from an empirical study evaluating **CASE**bot against a traditional worksheet, demonstrating that a genAI conversational agent is able to collect relevant contextual and preferential information, and combine it with an ideal structure to make self-experiment design more accessible by producing higher quality, better personalized, and safer experiment plans.
- Discussion around the tension between guidance and autonomy, leveraging a double-edged sword (genAI) to navigate the complexities of experimentation, and our vision for the role of genAI in making the self-experimentation practice at-large more accessible.

2 Related Work

2.1 Background on Self-Experimentation

Self-experimentation, also known as Single Case Designs (SCDs) or N-of-1 trials [21, 63], is an experiment design where an individual serves as their own control, highlighting their individual response to the intervention rather than a group's [63]. From 17th-century physician Santorio's experiments on bodily functions [66] to Nobel Laureate Barry Marshall's demonstration that Helicobacter pylori causes ulcers [41], self-experimentation has been critical to advancing our understanding of health and medicine [53, 79]. More recently, Neuringer and Roberts have popularized the approach as a tool to understand one's own personal health and wellbeing [72, 78].

For many years, analytical approaches for self-experimentation were limited to visual analysis, such as comparing before and after an intervention [72, 79]. Recent contributions by statisticians like Patrick Onghena have enabled more robust statistical tools for single-case designs, allowing randomized designs alongside traditional phase-based approaches [30, 42]. These advances, combined with automatic data collection via activity trackers [15], serve as catalysts for a self-experimentation movement [27, 30, 49]. From improving our understanding of orthopaedic surgeries [93] to our understanding of psychology [19], Weisse has identified 465 documented instances of self-experimentation in medical research in the past two centuries, with most of them conducted in the United States [96]. Self-experiments have empowered patients self-collecting data to understand treatments for cluster headaches [52]; to managing type 2 diabetes [38];

pain management [89]; exploring the potential in managing migraines [83, 84]; and understanding multiple sclerosis [4].

Despite its growing uptake, the practice of self-experimentation still remains largely inaccessible to the general population owing to it requiring expertise in experimental design, statistics, and specific domains. In this paper, we contribute to this movement by making the design of self-experiments more accessible, thereby enabling a broader range of people to conduct a wider range of experiments using their health data.

2.2 Self-experimentation in Personal Informatics

Given the focus of the quantified self movement on *knowing thyself* [65], self-experimentation has emerged as a tool for this community, and others who track themselves, to better understand their individual differences. Within the personal informatics research community, there has been growing interest in understanding and supporting the practice of self-experimentation [97]. In observing the practice, Choe et al. described a set of barriers that people run into when conducting such experimentation without support – not collecting the right data, undue data collection burden, and lacking scientific rigor [16]. Building on this understanding, systems like SleepCoacher and SleepBandits were created to help individuals experiment with sleep interventions [24, 25], TummyTrials was designed for discovering individualized triggers for irritable bowel syndrome (IBS) [47], Trialist focused on chronic pain [6], Galileo to support citizen science in gut health research [76], and explored the use of experimentation to support behavior change [58]. These systems addressed the barriers by developing bespoke designs that provide a guided structure to designing the self-experiment (e.g., step-by-step wizard to design the experiment), supporting personalization of experiment parameters (e.g., what time to collect data), and safety through involving domain experts. Such explorations have highlighted that designs can assist with self-experimentation, and also surfaced new challenges around leveraging technology to support the practice. Systems like TummyTrials and Self-E [23] addressed self-experimentation barriers by having domain experts pre-design experiments with customizable implementation (e.g., TummyTrials let users choose what foods they would have for breakfast foods). However, reliance on domain experts and pre-configured experiments creates a *scaling* problem. We lack technical mechanisms for supporting people in designing their own experiments while maintaining scientific rigor, support, personalization, and safety.

A self-experiment comprises three stages – design, execution, and analysis [48, 57]. Drawing from personal informatics literature [60], design is akin to the preparation stage, where individuals decide on hypothesis, variables, experiment design (e.g., phase-based A/B or randomized), and schedule [23, 48]. Execution is similar to collection stage, involves undergoing the experiment and collecting data [47]. Analysis resembles integration and reflection, examining data to evaluate the hypothesis [70]. An action stage often follows, where people decide on next steps based on findings. Following Rooksby et al.'s categorization, self-experimentation falls under a special category of self-tracking dubbed *diagnostic self-tracking* [80]. Consequently, self-experimentation shares properties of personal informatics systems where barriers cascade—mistakes in earlier stages carry forward and amplify [60], potentially resulting in failed or inconclusive experiments due to missing data, faulty design, or misleading findings [16, 48].

A common pitfall in experimentation occurs during the preparation stage, where the experiment design and planned data collection is not centered around an individual's broader health goals [32, 70]. Challenges arise during reflection when the collected data is too much or too sparse, lacks context, or is not appropriate for the user's question [8, 9, 20, 61]. This highlights the need for greater support in the design stage of the experiment to ensure a successful outcome.

CASEbot builds on these findings by starting with an individual's goal to better align the experiment; determining to track only the necessary data to reduce burden; taking personal context into consideration in the design; and scaffolding the scientific expertise. It does this while providing personalization based on the individual's context and constraints and safeguards against creating potentially harmful experiments.

2.3 Generative AI in Personal Informatics

Recent advances in genAI are influencing how we design technology to support personal informatics [3, 26, 44, 77, 90, 99]. Researchers use it to enhance personalization, making tailored recommendations for diet [39], sleep [92], and exercise [86] based on individual backgrounds and lifestyles [73]. However, as genAI increasingly addresses personal health questions [90], its non-deterministic nature raises safety and reliability concerns. These systems can hallucinate or give false advice confidently [62, 100], which could lead to harmful outcomes such as wrong recommendations to sensitive experiments and diminish user trust in health applications [95]. Assessing genAI trustworthiness is challenging, as these tools often lack clear credibility cues [29, 36]. Given these risks, using genAI for recommendations in sensitive health spaces like self-experimentation remains a significant challenge [45]. Our work addresses how **CASE**bot's structured, safety-conscious design can mitigate these concerns, providing a framework for responsible genAI use in this domain.

Recent work in personal informatics has explored genAI in contexts relevant to self-experimentation. In planning stage, *PlanFitting* engages users in dialogue to create personalized exercise plans framed as "if-then" intentions [86]. For data collection, *MindfulDiary* and *MindScape* use conversational journaling to document experiences and integrate passively collected data [33, 55, 71, 88]. *PHIA* and *GPTCoach* focus on analysis and reflection by helping users interpret health data by integrating personal and contextual factors [45, 69], *ExploreSelf* guides reflection on personal challenges through adaptive themes and Socratic questioning [87]. *HEALTHGURU* connects with wearables for context-aware sleep advice guided by behavior change theories [92]. Collectively, these systems demonstrate genAI's potential to support personal informatics—from planning and data collection to analysis—through personalized,

conversational interactions. Beyond personal informatics, research has examined how conversational agents can adapt to users for more effective collaboration. Studies show that agents are more engaging when they adjust to users' literacy levels and signs of confusion [31], match conversation length to question complexity [43], and tailor utterance complexity based on users' knowledge and prior understanding difficulties [1]. Our work with **CASE**bot builds on these explorations by applying similar techniques to self-experimentation. We structure experiment design as a step-by-step guide because decomposing complex tasks into discrete stages makes the process more manageable for users and enables the LLM to provide targeted support at each decision point. We adopt a coach interaction style to create supportive dialogue that helps users articulate their goals and constraints rather than overwhelming them with options. Finally, we ground our process in established self-experimentation frameworks to ensure scientific rigor and actionable insights.

3 Design

Our primary goal with **CASE**bot is to make the process of designing self-experiments more accessible, thereby empowering individuals to play a more informed role in their own health and wellbeing. As a first step in addressing the challenge, we seek to enable people to design personalized experiments which remain rigorous and safe to follow. We leverage genAI to design a system that can collect users' context and preferences and merge them with an ideal structure to support self-experiment design. We designed **CASE**bot to converse like a guide or a coach to increase the individual's involvement in the design process.

3.1 Design Principles

Prior research sought to address the three key barriers to self-experimentation – not collecting the right data, undue data collection burden, and lacking scientific rigor [16] – by designing platforms that scaffolded the necessary expertise through pre-designed experiments that provide a clear *structure* (i.e., step-by-step experiment setup), allow people a degree of *personalization* (e.g., deciding when to collect data), and ensured *safety* through involvement of domain experts in the experiment design [23, 47, 76]. However, such prior explorations were focused on supporting people with specific goals (e.g., identifying food triggering adverse symptoms) in choosing from a narrow set of pre-designed experiments. **If we want self-experimentation to be a more accessible practice, we need tools that allow individuals to create a wide-range of self-experiments that align with their goals.** Sticking with the above design principles, our aim with **CASE**bot is to allow people to conduct experiments beyond a small set of *templated*-experiments.

We designed **CASE**bot to ensure that experiments adhere to three core design principles:

- **Structure**, that ensures methodological rigor of the self-experiment.
- **Personalization**, which incorporates the user's lifestyle and constraints to promote engagement.
- **Safety**, to proactively check details to protect the user.

By operationalizing these principles through an interactive conversational flow, we aim to make the rigorous self-experimentation more accessible, enabling individuals to find evidence-based answers to personal health questions that they are interested in exploring.

We leverage prior literature in N-of-1 designs, particularly clinical science, medical informatics, and psychology [12, 48, 50, 57, 72] that inform us about the theory and design of self-experiments and effective goal-setting [28]. For instance, systems like TummyTrials [47] and Gut Instinct [75] demonstrated how to design structured experiment templates to guide users; Self-E [23], SleepBandits [25], and TummyTrials [47] demonstrated how to customize experiments to fit individual routines; and Galileo [76] showcased community vetting of suggestions to ensure the experiments remain safe and grounded. Leveraging these exemplars, we model how to structure the design of the self-experiment while allowing for some personalization and providing some measure of safety.

3.2 Structure

To ensure a scientifically rigorous self-experiment, **CASE**bot was designed to provide a structure by guiding the user through the four core dimensions articulated by prior work [48, 50, 57, 76]: Hypothesis formulation, Independent and Dependent Variable selection, appropriate Experiment Type and detailed Experiment Schedule in a linear, step-by-step manner (Prompt details in A.3).

Hypothesis. A well-defined hypothesis is crucial for a focused and meaningful investigation. However, prior literature has stated that self-experiment design benefits from starting with a discussion of the individual's goals [32, 70]. Towards this, **CASE**bot engages in a conversation to understand the user's broad health goals and the factors they believe are affecting them. The system has been prompted to gather details to collaboratively translate a person's goal into a hypothesis that is both testable and personally meaningful.

Independent and Dependent Variable. After hypothesis formulation, **CASE**bot collaborates with the user to specify experimental variables. Prior literature recommends that the *Independent Variable* (IV) be well-specified, controllable, and actionable, and the *Dependent Variable* (DV) be a quantifiable outcome reliably measurable within a defined post-intervention period [48, 50, 57]. Rather than suggesting generic interventions, the system discusses the user's current habits and daily routine. For example, for a dietary change, the agent asks about typical meals to suggest an IV that modifies an existing routine, such as adding a specific food to breakfast. It also confirms the intervention's feasibility. Finally, it ensures that the user has a reliable way to data collection –do they have a device that will automatically capture the data (e.g., step count) or if they require a spreadsheet template; and discusses measurement specifics (e.g., a 5-point scale for fatigue).

Experiment Type.* [1] After defining the variables, **CASE**bot guides the user in selecting an appropriate structure for the experiment. It prioritizes the method it infers to be most suitable based on the

[1]* To avoid overloading the phrase *experiment design*, what would typically be referred to as the experiment design is referred to as experiment type.

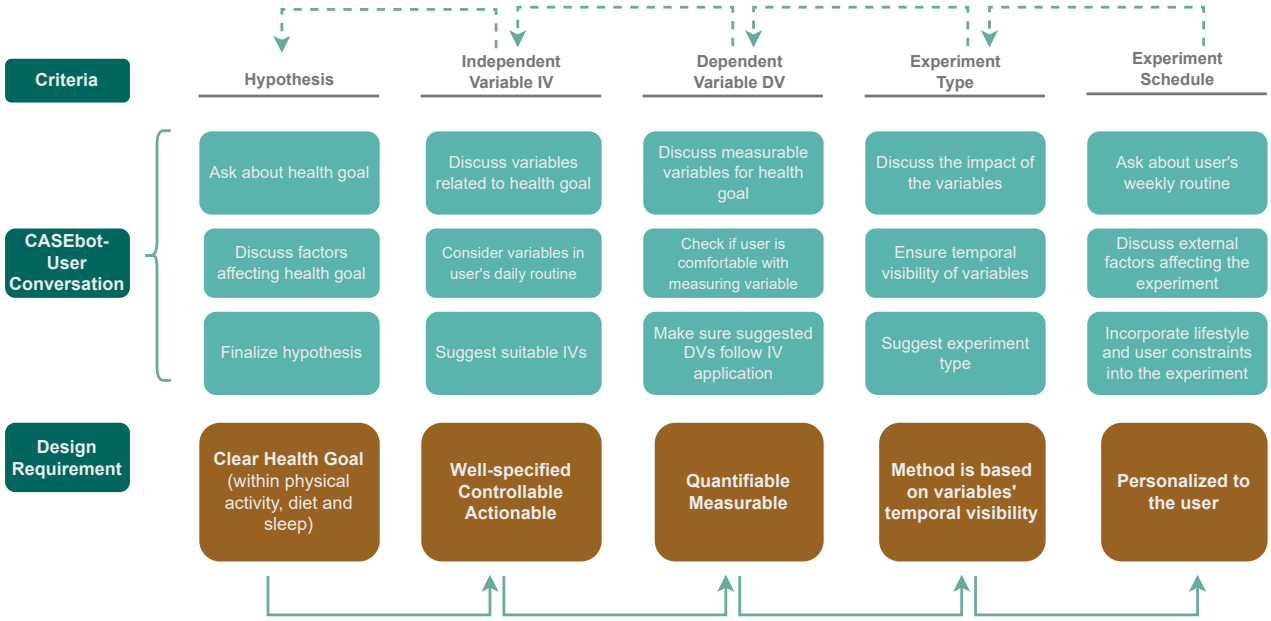

Figure 1: Design flow of CASEbot for creating valid self-experiments. The process consists of four dimensions of self-experimentation: Hypothesis, Variables: Independent Variable (IV) and Dependent Variable (DV), Experiment Type and Experiment Schedule. CASEbot-user conversation at each stage ensures that the system satisfies the corresponding Design Requirement. Solid line represents going forward whereas dashed line represents going backward as it is an iterative process.

temporal relationship between the variables, that is, how long it takes for the intervention's effects to become visible. The system will prioritize phase-based (A/B) designs, if they are better suited for interventions with delayed or "carry-over" effects [51, 57], which is common for experiments related to diet and sleep. Conversely, the system is instructed to suggest randomized designs only when the independent variable has a more immediate, short-lived effect [42, 74], ensuring the chosen experimental structure can validly detect a relationship.

Experiment Schedule. Prior work has shown that selecting an appropriate start date of the experiment is critical to reduce confounding variables (e.g., final exam week can influence mood) [16, 47]. The system was prompted to help build the schedule by asking about the user's daily life, habits, and specific times they are available (e.g., move the start date or recommend alternate experiment design). This ensures the final plan is integrated into their existing routine rather than being a disruptive or unrealistic commitment, thereby also potentially improving adherence.

3.3 Personalization

Personal informatics literature has demonstrated that to improve adherence to data collection and to provide value from the experience, the experience should be aligned to an individual's interests [9, 60, 61, 70]. **CASE**bot seeks to enhance personalization by learning about the user's preferences, daily schedule, and constraints (Additional prompt details in A.2 and A.5).

Context-Aware Variable Selection. Based on prior recommendations in behavior change and personal informatics, **CASE**bot aims to keep the data collection to be as low-burden as possible [8, 16, 60]. It achieves this in the following ways: For the Independent Variable, it suggests changes that are minor modifications to a user's existing habits (e.g., changing the timing of a meal they already eat) rather than introducing a demanding new behavior [37, 54, 78]. For the Dependent Variable, it works with the user to define an outcome metric that is not only valid but also one they are comfortable and capable of tracking consistently [48] (e.g., when tracking weight, suggesting to track it in the mornings to better integrate in daily schedule).

Realistic Intervention Selection. The system avoids generic intervention targets by basing the "dosage", or amount of change on the user's self reported baseline [34, 56]. For example, if a user wants to increase their water intake, the system first asks about their current consumption to suggest a realistic and incremental increase, ensuring the intervention is achievable for that individual.

Lifestyle-Integrated Scheduling. Based on prior literature, **CASE**bot seeks to design an experiment schedule to fit into the user's life [24, 25, 47]. By asking about specific daily and weekly commitments like work, classes, or social activities, the system aims to create a practical experiment calendar (schedule) that minimizes disruption.

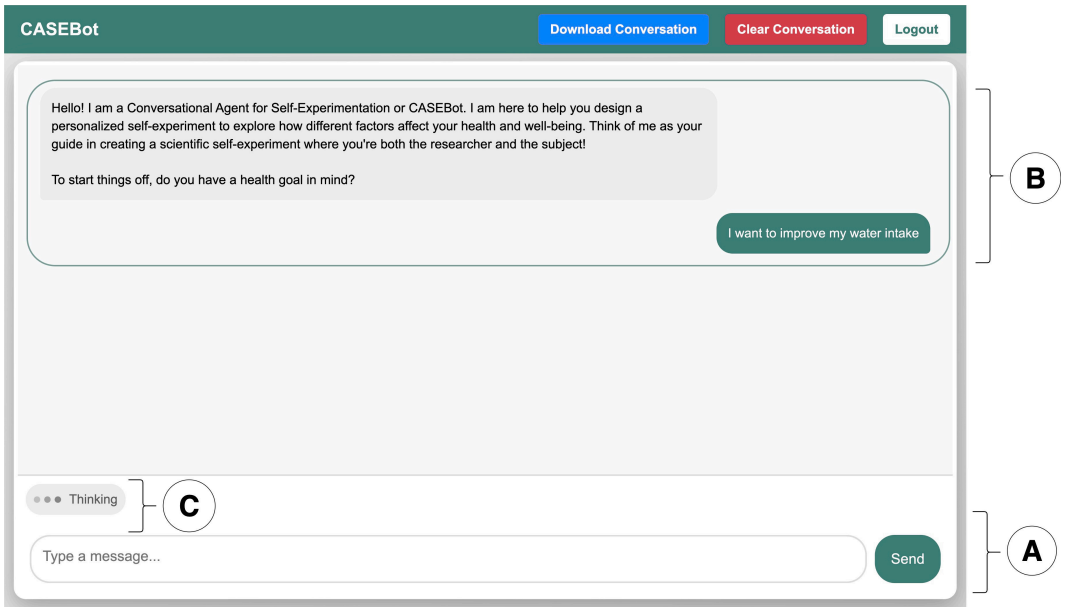

Figure 2: CASEbot's interface follows a typical chatbot structure for writing messages (A) and presenting responses as conversational dialog (B). A "Thinking" bubble (C) appears when CASEbot is communicating with the LLM to generate a response.

This ensures that the timing of interventions and measurements is feasible, to the extent possible.

3.4 Safety

A common concern regarding self-experimentation, particularly when it does not involve experts, is around its safety [54, 67]. Given that, safety has been a core principle of **CASE**bot. Prior research incorporated safety through involving domain-experts in the creation of the pre-designed self-experiments [23, 47]. However, that is difficult solution to scale. We also recognize that ensuring each experiment is safe for each individual requires high amounts of contextual information that is not available in our limited experimental setup. As such, our goal with **CASE**bot was to implement a *basic* safety standard wherein experiment interventions and dosages would be checked against reputed sources (e.g., Centers for Disease Control and Prevention, U.S. Food and Drug Administration and National Institutes of Health) for being *reasonable* [14, 91]. **CASE**bot's proactive approach to safety is implemented through two mechanisms:

Screening Potentially Unsafe Hypotheses and Goals. A top-level safety mechanism that **CASE**bot implements is to restrict the self-experimentation to domains that the research team felt confident in evaluating without needing any special expertise. This meant that the experiments were limited to the domains of physical activity, diet, and sleep. The process begins by ensuring the user's health goal is appropriate for an experiment – the goal is feasible and realistic based on their lifestyle and should also adhere to the three specified domains . The system asks for relevant background health information early in the conversation, allowing it to identify potential risks before a hypothesis is even finalized (as described in A.4.1).

Verifying Variables. The system forbids from allowing users to choose an IV related to pharmacological drugs or psychoactive substances (besides caffeine). In addition, when an intervention involved dosage (e.g., caffeine), the system is instructed to check the proposed amount against established recommendations from sources like the CDC and FDA to ensure it is within safe limits (as described in A.4.2). It would also avoid suggesting a DV that requires a physically demanding or potentially harmful measurement process for an individual with certain health conditions. Questions about lifestyle and habits also helped with dosage limits and avoiding creating schedules that could lead to burnout or physical strain (as described in A.4.5).

3.5 System Implementation

We refined **CASE**bot's design through iterative prompt engineering. Our initial prompts defined self-experimentation and core dimensions, but early interactions with the system lacked the interactivity we desired, instead providing direct answers (e.g., exactly what experiment to run) rather than guidance. We addressed this by adding collaborative dialogue instructions to guide users through the process. To support the LLM in navigating relevant health domain related suggestions, we enhanced the prompt with three detailed domain-specific examples. We developed these examples by borrowing example self-experiments from prior literature in the domains of physical activity, diet, and sleep [22, 48, 78] and applying our design principles – structure, personalization, and safety (Full list of

examples in A.6). While we acknowledge this scope, prior work on self-tracking and the quantified self consistently identifies physical activity, diet, and sleep as the primary domains where individuals conduct behavioral self-experiments [35]. Furthermore, our design principles are generalizable, and future work could extend CASEbot to additional domains by incorporating appropriately validated examples.

The system uses the Rasa framework [10] with two modules: Natural Language Understanding (NLU) and Response Generation, both powered by Anthropic's Claude 3.7 Sonnet (temperature 0.4, max_tokens 4096) [2]. At the time of our testing, despite our best attempt to prompt the models not to do so, Google's Gemini and OpenAI's ChatGPT 4.0 frequently provided one-shot answers rather than conversational exchanges, while Claude was more consistently able to perform multi-turn dialogues (e.g., in at 80% of our tests with self-experiments in physical activity, diet and sleep). The NLU module uses in-context learning to classify user intent, while response generation retrieves conversation history from MongoDB and prompts the LLM to generate responses that guide users through the design process or redirect off-topic conversations. Figure 2 shows the **CASE**bot interface used in our study.

3.6 Worksheet Design

A platform supporting domain-agnostic self-experiment design would be an ideal comparison to **CASE**bot. However, we were not aware of such a tool, existing tools are aimed at supporting administration of pre-designed experiments rather than custom experiments [23, 47, 82] and are often unmaintained post-publication. We therefore leveraged pen-and-paper self-tracking methods [5, 16, 60] to develop a comparable artifact.

We therefore designed and compared against a traditional worksheet to aid the design of self-experiments. This digital worksheet was constructed to serve as a non-interactive but structurally equivalent counterpart to **CASE**bot, requiring users to define the same five criteria for their experiment: hypothesis, independent variable, dependent variable, experiment type, and experiment schedule. The criteria were elicited in the form of open-ended questions (e.g., "What is your hypothesis?", "What is the independent variable? a. How will you collect it? b. How often will you collect it?"), but requiring the user to make experiment design decisions independently. It afforded personalization by asking people to think about their routines and preferences but it was not done interactively. The use of a worksheet allowed us to isolate the impact of the conversational guidance, feedback, and personalization offered by **CASE**bot, rather than just the effect of being presented with a structured template. Participants also had access to a slide deck with a refresher on experiment design and experimentation in case they needed to refer to any specifics or terminology (see Supplement Materials for a copy of the worksheet and the slide deck).

4 Methods

We conducted a within-subjects, mixed-methods study to evaluate the feasibility of **CASE**bot in supporting users towards designing well-structured, personalized, and safe self-experiments. Figure 3 depicts the study workflow.

4.1 Study Procedure

The study was conducted as part of an assignment in a health informatics course at a large university in the US. Given the primary emphasis of **CASE**bot is to increase the variety of experiments, it was convenient to recruit students in a class. Prior research in self-experimentation has adopted similar recruitment strategies to understand self-experiment design [22]. Being mindful of concerns regarding coercion to participate, we worked closely in collaboration with our University's Human Subjects and Research Engagement (Institutional Review Board, IRB#6353) to design the protocol wherein appropriate steps were taken to ensure that students were not pressured to participate in the research. At the time the students were presented with the assignment, prior to any form of data collection, they were made aware of our intent to use their data for a research study. Students were also provided an overview of how research participation works in general, what rights participants have, and were introduced to the consenting process we would adopt for our research. They were given an opportunity to ask questions about the study during the class and were walked through the entire timeline depicted in Figure 3. In accordance with the IRB's guidance, student consent was only collected once the assignment had been graded and the grades were locked and published (i.e., visible to the students). This allowed students (henceforth referred to as participants) to provide a meaningful consent without worrying about their decision impacting their grade.

Prior to week 1 of the study, participants received an introduction to the core components of scientifically valid self-experiments. The lecture covered the idea of self-experimentation, how to formulate a good hypothesis, how to select appropriate variables, and different experiment types. For both weeks, participants were also provided with a brief refresher (summary of the previous lecture) prior to them designing the experiment, that covered the basics and the terminology (see supplement material for slide deck). Participants were divided into two equal sections and were asked to design two self-experiments, one using **CASE**bot and another using worksheet, a week apart. For the first week, section 1 used **CASE**bot and section 2 used worksheet. For the second week, the platforms were switched, to help control for ordering effects. Participants were instructed not to repeat the same experiment design across the two week/sessions. Due to class constraints, both sessions were limited to ~50 minutes within which participants had to design the experiment plan and submit the survey response, which is in line with a typical lab-based research study.

For each session, participants spent the majority of their time designing their self-experiment using the assignment platform. During this time, if they were assigned to the **CASE**bot section, it recorded their conversational log. Once they were done designing the experiment, they had to fill out a survey to capture their experience and feedback on the platform (See supplement material). The survey consisted of four parts: 1) share their self-experiment design; 2) likert-like questions about their impression of the platform, their perspective on how the platform support the experiment design process, and effectiveness of the platform; 3) open-ended qualitative questions focusing on

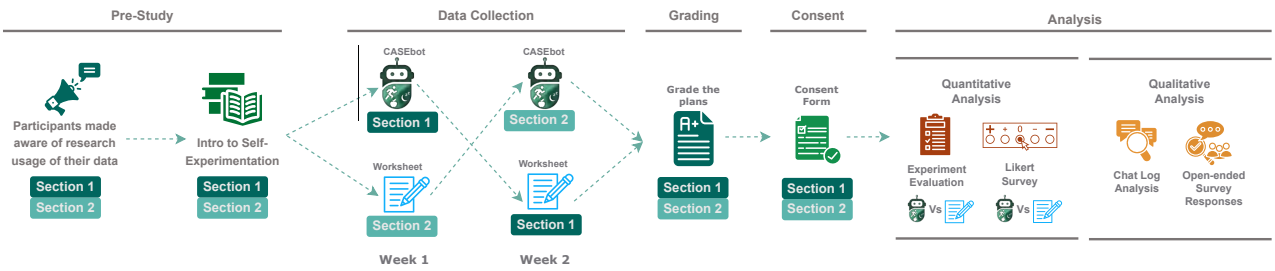

Figure 3: An overview of the study procedure. All participants were divided into two sections: Section 1 and Section 2. All participants received an overview of the study plan and an introductory lesson on self-experimentation before the study took place in two consecutive weeks. In first week, Section 1 used the CASEbot, while the other used the worksheet, and then they swapped in the second week. Consent was collected only after the grades for the assignment were published to avoid any coercion. Consented participants' data were used for analysis which involved both quantitative and qualitative methods.

participants' reflection on their process and platform; and 4) basic demographic and relevant experience information.

Once both weeks of the study concluded, instructors graded the assignments as part of routine coursework. Once the grades were released, participants received an online consent form asking for their permission to use the data for the research study. To further ensure the consent was well-informed, the form explicitly asked for permission to use their assignment, **CASE**bot chat transcript, and their reflection form (see consent in supplemental material). Only participants who consented to provide access to all three data sources were included in our analysis. All data – conversational logs, experiment designs, etc. – of the participants who did not grant consent or did not respond to our request were deleted.

All data used for analysis were fully anonymized. When interacting with the Claude API, only anonymized text was sent, ensuring no personally identifiable information was exposed. Participants were not compensated for their participation.

4.2 Participants

All of the participants were undergraduate students (juniors and seniors). Of the total of 52 students enrolled, 43 participants provided consent to have their data be used in our research analysis. One participant submitted the same experiment plan for both **CASE**bot-guided and worksheet-guided one, and we removed data from the analysis and did not report on their demographics. This gave us a final count of 42 consented participants (21 in section 1 and 21 in section 2).

Out of 42 participants, 21 self-identified as female, 19 as male, and 2 as non-binary. 37 participants were informatics majors, four were computer science majors, and one data science major. 38 participants did not have prior experience with self-experimentation, 3 had previous experience and 1 was unknown. Conversely, 36 participants reported maintaining some kind of health-related self-tracking. All participants reported having taken a statistics course, either in high school, college, or both. For hypothesis testing and statistical evaluation of experiments, 34 participants reported to be comfortable, 5 being Neither comfortable nor uncomfortable and 2 participants being uncomfortable.

4.3 Experiment Evaluation

It is difficult to objectively compare the quality of self-experiments which have different hypotheses, variables, and schedules. Moreover, comparing self-experiment designs across participants is not a common evaluation goal in prior work. As such, we sought to define a rubric that would allow us to compare the design of user-generated self-experiments. We developed this scoring rubric drawing from prior literature on self-experimentation and behavior change, with a focus on whether experiments were specific, actionable, and safe [28, 48]. While we did not conduct a formal evaluation, we verified that self-experiments from prior systems with expert involvement would score highly. For example, expert-guided self-experiments have been clear in indicating weekly schedules with control and intervention dates, such as ABAB or randomized designs; allowing people to choose and personalize their experiment parameters; and leverage experts to ensure experiment designs are safe [23, 24, 47].

This rubric is structured around the five key criteria of a self-experiment — hypothesis (4 points), independent variable (5 points), dependent variable (4 points), experiment type (6 points), and schedule (4 points) — for a total of 23 points. Each criterion is further broken down into specific sub-criteria (see Table 1). **This scoring system allowed for a direct comparison between plans created using CASEbot and the baseline worksheet**

4.3.1 *Hypothesis (4 points).*

- Health goal (1 point): Is hypothesis driven by a health goal?
- Relationship between variables (3 points): A well-defined hypothesis must explicitly state the relationship between the independent and dependent variables [40, 48]. This is broken down into whether it mentions the IV (1 point), the DV (1 point), and the relationship between them (1 point).

4.3.2 *Independent Variable (5 points).*

- Specific (4 points): Is the dosage/amount clearly mentioned (2 points)? Scored as: 2 points for a specific dosage and variable, 1 point if the variable is chosen but the dosage is missing, and 0 point if neither. If the timing of the intervention is

Table 1: The evaluation rubric for assessing the quality of self-experiment plans. The rubric consists of five main criteria, each with multiple sub-criteria, for a total of 23 points.

Criteria	Sub-criteria	Description	Points	Total
1. Hypothesis	**1(a)** Driven by Health Goal	Whether it relates to a health goal that the experimenter had in mind.	(0/1)	4
	1(b) Mentions IV **1(c)** Mentions DV **1(d)** Mentions relationship	It clearly states IV and DV and their relationship.	(0/1/2/3)	
2. Independent Variables	**2(a)** Specific: Dosage (Measurable)	Dosage is mentioned.	(0/1/2)	5
	2(b) Specific: Time	The IV measurement is scheduled for a specific time.	(0/1)	
	2(c) Feasibility	The IV is administrable reliably.	(0/1)	
	2(d) Controllable (and by extension -Actionable)	The variable is under the control of the experimenter.	(0/1)	
3. Dependent Variable	**3(a)** Specific: Consistency	Is the DV measured in a consistent manner?	(0/1)	4
	3(b) Specific: Time	The DV measurement is scheduled for a specific time	(0/1)	
	3(c) Feasibility	The DV is administrable reliably.	(0/1)	
	3(d) Quantifiable	Quantification criteria are mentioned.	(0/1)	
4. Experiment Type	**4(a)** Temporal visibility	Is the effect of IV on DV possible to observe in the scheduled window?	(0/1)	6
	4(b) Suitable	Randomized/phase-base.	(0/1)	
	4(c) IV:Safety	Deduce Safety from the specificity sub-criteria of IV.	(0/1/2)	
	4(d) DV:Safety	Deduce Safety from the specificity sub-criteria of DV.	(0/1/2)	
5. Experiment Schedule	**5(a)** Weekly schedule	The experiment schedule is detailed enough to act on.	(0/1)	4
	5(b) Starting date	Mentions the starting date.	(0/1)	
	5(c) Control days mentioned	Control days are chosen correctly.	(0/1)	
	5(d) Intervention days mentioned	Intervention days are chosen correctly.	(0/1)	
				= 23

defined (1 point) and if the proposed intervention is realistic and doable for the user in their daily life (1 point) [50].

- Controllable (1 point): Can the user easily manipulate the variable [50]?

4.3.3 *Dependent Variable (4 points).*

- Specific (3 points): Is the variable measurable with consistency (1 point), at specific times (1 point), and is realistically feasible (1 point) to track [50]?
- Quantifiable (1 point): Experiment includes clear quantification criteria [50].

4.3.4 *Experiment Type (6 points).*

- Appropriateness (2 points): Temporal visibility or whether the effect of the IV on the DV is observable within the scheduled window and its suitability (1 point). Is the experiment type (e.g., randomized or phase-based) suitable for the variables chosen (1 point) [46, 57]?
- Safety (4 points): Safety evaluation for both the IV (2 points) and the DV (2 points), ensuring the experiment does not pose unnecessary risks. Safety is rated on a 3-point scale (unsafe, not rateable, safe). A plan is deemed 'not rateable' if it lacks the specificity needed to assess potential risks, such as a missing dosage for a supplement or unclear measurement procedures [46, 57].

4.3.5 *Experiment Schedule (4 points).*

- Detailed (4 points): Is a weekly schedule provided (1 point)? Is a specific starting date mentioned (1 point)? And, are the control day (1 point) and intervention day (1 point) conditions highlighted in sufficient detail [48]?

4.4 Analysis

The study consisted of three data sources: the design of the **CASE**bot and worksheet-guided self-experiments, survey responses following both weeks, the transcript of each **CASE**bot conversation. We quantitatively analyzed the self-experiment designs based on their rubric scores and participant responses to the likert-like questions in the survey. We qualitatively analyzed participants open-ended responses in the survey as well as the chat transcripts.

4.4.1 Quantitative Analysis. Each self-experiment plan was evaluated based on the evaluation metric described in section 4.3. The first two authors collaboratively evaluated the plans by dividing the data in half, with each author reviewing one portion,

and then switching to ensure consistency across evaluations. One of the authors conducted all the quantitative analyses.

To statistically analyze the experiments, we used linear mixed effects model on each sub-criterion, considering the platform (**CASE**bot and worksheet) and section order (Section 1 and Section 2) as fixed effects and participants (PIDs) level variability as random effect. Across all the linear mixed effect model analyses, the random effect variances were effectively zero/nearly zero, suggesting a consistent participant-level performance. Hence, we did not report the random effect variance in the detailed results. Additionally, results indicated that there is no significant effect of ordering (Section 1 versus Section 2). As a result, we disregarded this fixed effect. We reported the average score difference between the **CASE**bot and worksheet group indicated by the estimated coefficient (Coef), along with the Z-value, p-value and 95% CI.

4.4.2 Qualitative Analysis. We open coded the participant's responses around their experience on using the platforms. We applied thematic coding following the structure defined in [11] and the execution demonstrated in [13]. The first two authors conducted the initial qualitative analysis by dividing the data according to the two study sections. First, the authors familiarized with the data, and then proceeded to identify certain codes, and finally forming themes out of the codes. Following this, one of the first authors reviewed all data and developed the thematic structure. After initial coding and discussion with the research team, we settled on coding participant experiences according to **CASE**bot's key system features of structure, personalization, and safety, labeling positive and negative experiences with both **CASE**bot and worksheet towards those features. Following this analysis, we also analyzed the chat transcripts of the participants to look for certain trends in the conversation that demonstrated positive and negative examples of **CASE**bot regarding structure, personalization and safety.

4.5 Limitations

We limited the self-experimentation to three domains which the research team felt comfortable evaluating for safety - physical activity, diet, and sleep. Although we designed **CASE**bot to be domain-general, further evaluation is needed to demonstrate the effectiveness of the approach for other experiment domains, such as mental health. We acknowledge that participants' potentially higher level of digital, especially AI/LLM, literacy–as Informatics and Computer Science majors–may have shaped their interaction with **CASE**bot. Their familiarity with experimental design and statistics may also have afforded them more autonomy in engaging with the system, including critically evaluating, modifying, or discarding suggestions made by **CASE**bot. This might not reflect interaction patterns of less technically experienced users.

Additionally, the study took place in a classroom setting, and participants' behavior might have been shaped by their concern for evaluation in the academic context. Some participants prioritized having a complete self-experiment plan submitted (e.g., in order to secure good assignment grade) rather than deeply engage with the chatbot's guidance. A few participants asked verbatim reflection survey questions regarding certain experiment components , which both indicated that they were likely not deeply invested in their self-experiment topic and likely hindered the process of building an incrementally structured and personalized plan. Further, each study session was limited to ~50 minutes during which participants had to spend some time on familiarizing with the system and study protocol. This might have prevented the participants from engaging with the system at their own pace. Finally, based on the lifestyles of students, the types of experiments that students proposed were likely more limited and specialized relative to the general public. It is unclear whether an LLM-based approach may be more or less adept at supporting these types of experiments compared to others. For example, given societal conversations around ensuring students get enough sleep, LLMs might have ample examples from online for how to support students in getting quality sleep. This could lead to higher-quality experiment design compared to other population's sleep goals.

CASEbot's design is a proof-of-concept and as such has its own limitations in adhering to our broader design principles (section 3.1). Despite our prompts asking it not to provide direct answers but to instead work with the participants, we saw some examples of it inadvertently doing so, although participants were still able to edit or ignore those responses. As an LLM, that is parsing user-generated text for establishing context, **CASE**bot was not particularly effective at personalizing experiment designs when participants gave fairly curt responses. Additionally, while we took a few measures to ensure experiment safety, our system was fairly limited in its ability to adhere to this principle. For example, **CASE**bot was not designed to point out safety risks which may be related to a person's prior history or patterns, such as discouraging someone who has no history of cardio activity from planning to run a 10k every day. Because such a behavior is potentially safe for someone with experience, it is unlikely to be flagged by reputed sources. Better understanding what experiment set-up is reasonable and safe for an individual is therefore a critical opportunity for future work.

5 Results

In this section, we illustrate findings from our study with a focus on our three design principles: structure, personalization, and safety. Throughout the analysis, we considered subjective experiences reported by the participants to support the quantitative findings.

5.1 Overall Performance and Experience

When formally comparing the experiment rigor and specificity using the rubric, most participants designed better experiments using **CASE**bot. **CASE**bot-guided plans received an average score of 22.31 (out of 23) whereas worksheet-guided plans received 18.98, with an average score improvement of 19.20% with **CASE**bot within the participants. Overall, as shown in Table 2, participants reported greater **satisfaction** with the experiments they created with **CASE**bot rather than worksheet (O1: Z=2.237, p=0.025, 95% CI 0.056-0.849 on a 7-point scale). Participants also rated **CASE**bot more highly than worksheet on most other dimensions – ease of use (O2), reasonableness of experiment (O3), time taken (O4), confidence in design (O5), learning from process (O6), and preference (O7). We observed no statistical difference about people's perception of their ability to design better experiment

Table 2: Results for Survey Questions on Participant's Overall Experience on Quality and Process of the Experiment (on a 7-point scale): Participants generally found CASEbot easier to use, and were more satisfied with the experiments they created.

	Survey Question	CASEbot Effect	[95% CI]	z-score	p-value
O1:	Overall, I am satisfied with the self-experiment that I designed using the [platform]	0.452	[0.056, 0.849]	2.237	0.025
O2:	It was easy to design a self-experiment using the [platform]	1.024	[0.568, 1.479]	4.406	< 0.001
O3:	The questions the [platform] asked to design the experiment were reasonable	0.357	[0.091, 0.623]	2.635	0.008
O4:	The amount of time I took to design a self-experiment using the [platform] is reasonable	0.452	[0.131, 0.774]	2.756	0.006
O5:	I am confident that I can successfully collect the data for the self-experiment based on the planned design	0.452	[0.104, 0.801]	2.545	0.011
O6:	I learned how to design good self-experiments by using the [platform]	0.762	[0.379, 1.145]	3.903	< 0.001
O7:	If I were to design a self-experiment in the future, I would prefer using the [platform]	0.976	[0.314, 1.639]	2.888	0.004
O8:	I think I could design a self-experiment that is better than the one the [platform] designed	0.310	[-0.278, 0.898]	1.032	0.302

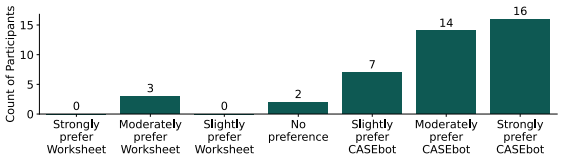

Figure 4: Participants' perceived preference for self-experimentation platform

than the platform (O8: $p = 0.302$). Finally, we observed a clear preference for **CASE**bot among the participants – 37 would prefer **CASE**bot versus only 3 who preferred worksheet (see Figure 4).

Participants appreciated worksheet's clear focus on the important aspects of the experiment design and the flexibility it provided. However, a critical limitation they identified was the lack of guidance and feedback in the process. On the other hand, participants appreciated **CASE**bot's conversational approach, specifically its step-by-step guidance in designing the self-experiment. They also reported it personalized the designs by prompting them to surface everyday constraints and integrating them in the plan. Finally, looking at the chat transcripts, we found that **CASE**bot proactively raised safety concerns leading to safer experiments. However, some participants did not appreciate the verbose communication and found it rigid in its recommendations.

5.2 Adherence to Structure

Overall, participants perceived the worksheet as supporting structure through its organization which offered clear instructions along the way whereas **CASE**bot provided structure through its strong and effective guidance and specificity of its recommendations. Although the worksheet provided the structure for the major components of a self-experiment plan, participants perceived it lacking in providing guidance to develop one. Participants considered the guidance provided by **CASE**bot to be effectively organized across the structural criteria as shown in Figure 5. According to participants' perception on structured guidance of both platforms, **CASE**bot-supported plans scored higher than worksheet ones (Z=3.719, p<0.000, 95% CI 0.282-0.909 on a 7-point Likert scale). Figure 6 shows the percentage of maximum of the aggregated score in each individual criteria for both platforms. Below, we focus on the support provided by **CASE**bot and worksheet towards generating well-structured plans across the four key criteria.

5.2.1 ***Hypothesis.*** When aggregating scores across Hypothesis sub-criteria, we observed no statistical differences between Self-experiment plans guided by **CASE**bot and worksheet (p=1.000).

Although there was no statistically significant difference between the platforms, participants' perspective painted a different picture. As in Figure 5, participants perceived **CASE**bot to be more effective in *Generating hypothesis* (Z=4.859, p<0.001 and 95% CI 0.767-1.804 points out of 7). Participants appreciated that both **CASE**bot and worksheet separated out hypothesis formulation into an explicit step. As P42 said, "*It [worksheet] is pretty concise and clear to me and helps me understand my goals better in my experiement.*" However, participants also found the leap from health goal to a hypothesis to be challenging. As P13 stated, "*going straight from what is your health goal to what is your hypothesis was not very helpful. I wish there were a few more questions*".

Conversely, participants appreciated **CASE**bot's follow-up questions and conversational refinement as clarifying and motivating. P50 reflected: "*The first moment I was truly impressed by **CASE**bot was when it came up with the hypothesis...This is very thorough. At first, I was confused why **CASE**bot was asking all these questions, but it was in this moment [creating the hypothesis] where I realized how necessary and important it was.*" Similarly, P48 said: "*I think it's a great way to get started with forming your hypothesis. It also asks interesting questions that help you think outside the box.*".

Analyzing the chat transcripts further revealed **CASE**bot's capability in effectively breaking down a health goal into specifics. Figure 7 illustrates one such example. P36 wanted to experiment on how hydration affects endurance. With that as the starting point, **CASE**bot built towards a specific hypothesis by eliciting participant's details about existing activity routine (on strairmaster) and context (on hydration). As the participant supplied concrete schedule and baseline behavior, **CASE**bot incorporated it to set timings for baseline and intervention. Finally **CASE**bot formulated a clear and testable hypothesis.

5.2.2 ***Variables: IV and DV.*** As shown in Figure 6, plans co-designed with **CASE**bot scored higher in both Independent and Dependent variable criteria than the plans created with worksheet [(Z=6.562, p<0.001 and 95% CI 0.701-1.299 points out of 5) and (Z=4.635, p<0.001 and 95% CI 0.522-1.287 points out of 4) respectively]. Table 3 shows analyses on each of

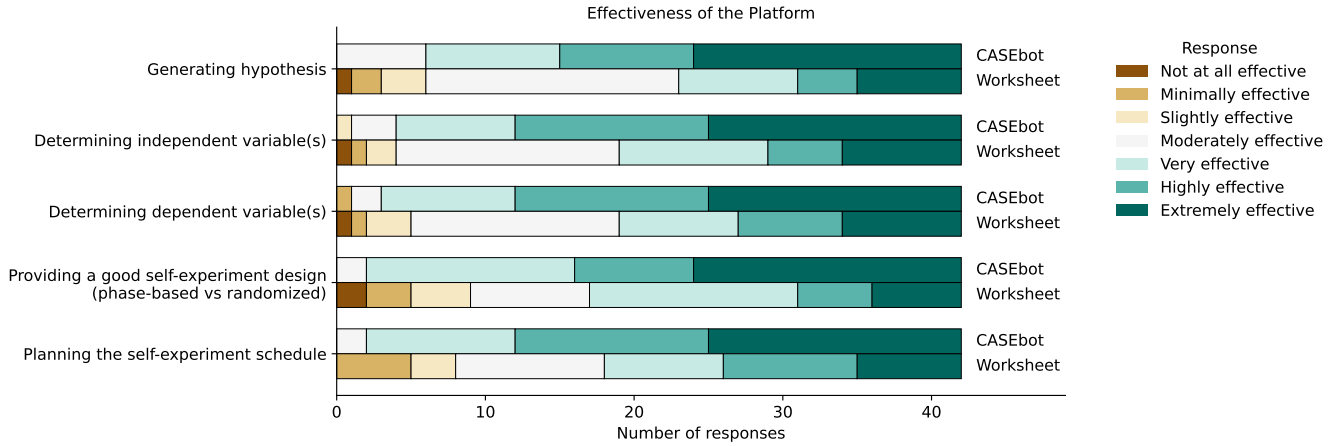

Figure 5: Participants' perception of effectiveness of both platforms in each step of planning

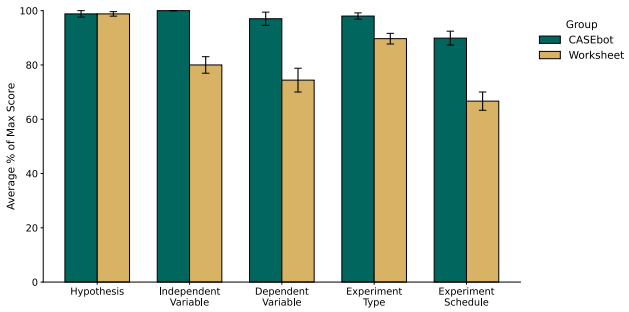

Figure 6: Average % of maximum score comparison between CASEbot-supported and worksheet-supported plans across 5 key criteria

the 4 sub-criteria of IV (**2(a)** *IV:Dosage*, **2(b)** *IV:Specific-Time*, **2(c)** *IV:Specific-Feasibility*, **2(d)** *IV:Controllable*) and DV (**3(a)** *DV:Consistency*, **3(b)** *DV:Time*, **3(c)** *DV:Feasibility*, **3(d)** *DV:Measurable*) along with their aggregate results (see Section 4.3).

CASEbot-supported plans scored higher on **2(a)**(Z=5.266, p<0.001, and 95% CI= 0.359-0.784 out of 2 points) and **2(b)** (Z=4.773, p<0.001, 95% CI 0.210-0.504 out of 1 point), indicating the inclusion of specific independent variable in the plan. We did not observe significant differences with **2(c)** (p = 0.152) and **2(d)** (p = 0.317) sub-criteria. **CASE**bot-guided plans scored higher than worksheet-guided ones on **3(d)** (Z=4.343, p<0.001, 95% CI 0.196-0.518 out of 1 point), **3(a)** (Z=3.276, p=0.001, 95% CI 0.096-0.381 out of 1 point), and **3(b)** (Z=3.517, p<0.001, 95% CI 0.116-0.408 out of 1 point). We did not observe a significant difference in **3(c)** (p = 0.308). These results suggest that, while both the platforms were able to produce relevant and meaningful dependent variables, **CASE**bot guided the participants towards defining more appropriate and practical DVs. As in Figure 5, participants perceived **CASE**bot to be more effective in *Determining independent variable(s)* (Z=4.634, p<0.001 and 95% CI 0.646-1.592 points out of 7) and *Determining dependent variable(s)* (Z=4.554, p<0.001 and 95% CI 0.624-1.567 points out of 7)].

Table 3: Summary of Results for Variable (IV and DV) Criteria: With the exception of 2(c), 2(d), and 3(c), CASEbot-guided plans outperformed the worksheet with statistically significant improvements across all sub-criteria.

Criteria	Group: CASEbot [95% CI]	z-value	p-value
2 Independent Variable Aggregate	1.000 [0.701, 1.299]	6.562	< 0.001
2(a) IV: Dosage*	0.571 [0.359, 0.784]	5.266	< 0.001
2(b) IV: Specific-Time	0.357 [0.210, 0.504]	4.773	< 0.001
2(c) IV: Specific-Feasibility	0.048 [-0.018, 0.113]	1.432	0.152
2(d) IV: Controllable	0.024 [-0.023, 0.070]	1.000	0.317
3 Dependent Variable Aggregate	0.905 [0.522, 1.287]	4.635	< 0.001
3(a) DV: Specific–Consistency	0.238 [0.096, 0.381]	3.276	0.001
3(b) DV: Specific–Time	0.262 [0.116, 0.408]	3.517	< 0.001
3(c) DV: Specific–Feasibility	0.048 [-0.044, 0.139]	1.019	0.308
3(d) DV: Measurable	0.357 [0.196, 0.518]	4.343	< 0.001

**Note:* IV criteria is on 5-point scale and DV criteria is on 4-point scale. All sub-criteria are binary (0 or 1) except for 2(a) which is on a 2-point scale (0, 1, or 2)

11 participants mentioned that worksheet was particularly helpful towards identifying the variables. As P47 reflected on worksheet: "*I think explaining the independent and dependent variable helped me understand what I needed to change and what I was actually measuring.*" Although the worksheet introduced participants to how plans should be structured, participants found it lacking in guidance towards steps like identifying what variable to test. As P8 mentioned, "*I believe it's very hard for beginners who need more guidance on variables and analysis plans if they are only provided with the worksheet*". Similarly, P46 remarked, "*[For] worksheet, I had to think more about the independent and dependent variables along with actually figuring out issues when it came to designing it. While the **CASE**bot it was easier since it could just design it.*". Participants were often unsure where to start with selecting variables, or even what was meant by the terminology, which was a barrier to using worksheet. P20 said, "*However it [worksheet] lacked sufficient instructions on what the*

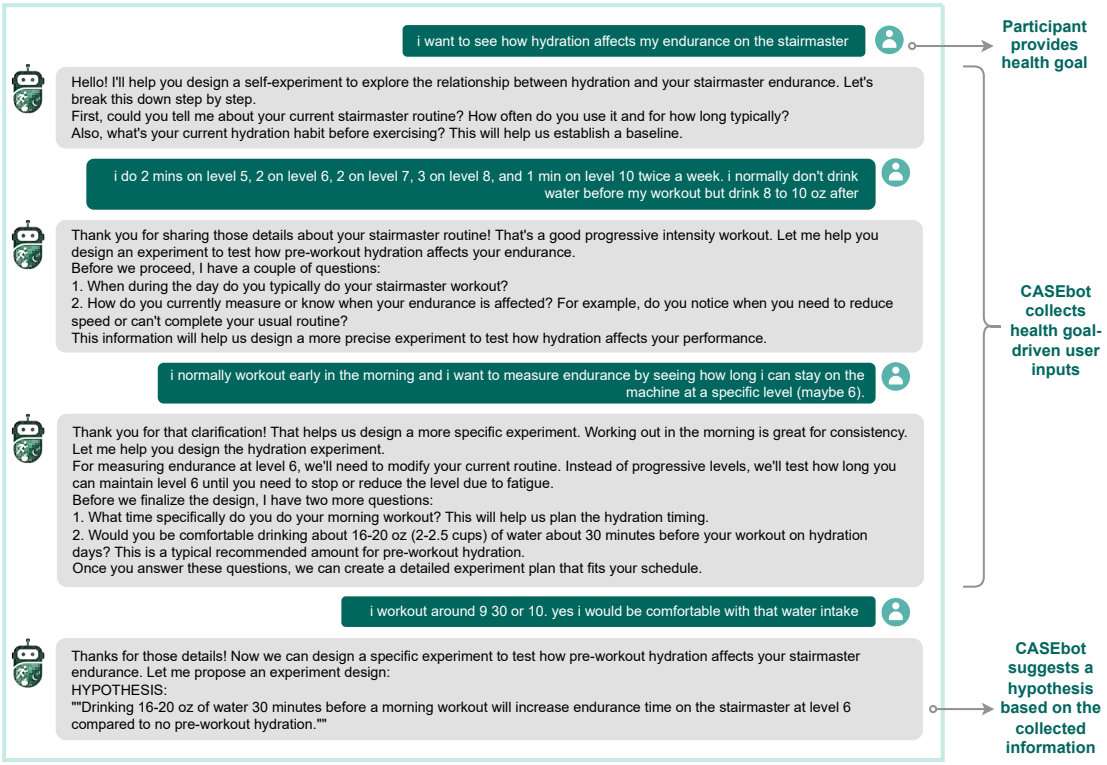

Figure 7: Chat transcript excerpt of P36 on how hypothesis is formulated.

independent or dependent var means and what analysis should entail". Similarly, P45 said, "*I was not exactly sure what would be dependent or independent variables without looking at the lecture slides*". Participants reported having trouble evaluating whether their dependent variables would lead to valid or useful experiments. P13 said, "*Using the worksheet was much more difficult than using the ChatBot [**CASE**bot]. I had to change my dependent variables a few times because I wasnt sure if they were good enough.*".

In contrast, participants found the probing strategy of **CASE**bot to be helpful in identifying variables. P12 said, "*it [**CASE**bot] asks about my health goals, but it also goes deeper asking specifics about my sleep schedule like what affects it and how I sleep. And then it latches onto a good independent variable and develops the self-experiment from there.*". Participants found that **CASE**bot used their input to generate suggestions which were beyond what they could have developed on their own. P33 said, "*A lot more in-depth analysis of my patterns and use of my resources. I could not consider points such as wind-down activity or come up with such robust variables without the help of **CASE**bot*". Participants appreciated **CASE**bot's guidance in establishing quantifiable intervention targets. P44 mentioned, "*a particularly helpful interaction was when we established clear measurement points for energy levels (11 AM, 4 PM, and 7 PM) and defined the specific water intake amount (333ml). This made the experiment more precise and actionable, rather than just having a vague goal about drinking more water and tracking energy.*".

Participants also described some cases where **CASE**bot failed to incorporate their preferences on variables to track when they had gone further into the structured design process. P8 said, "*I wanted to track meditation and my freshness upon waking but also account for having inconsistent bedtimes. I hoped to add this variable to my existing experiment but **CASE**bot suggested creating a new experiment instead.*". Upon analyzing the transcript to understand the nature of the issue, we discovered that P8 brought up the requirement around bedtime after **CASE**bot had designed a plan and instead of editing the designed plan, **CASE**bot reverted to an earlier step to ask further questions which ended up confusing the participant.

Analyzing the variable identification (IV or DV) phase from the chat transcripts showed that **CASE**bot was generally able to support participants in turning their abstract goals into specific and feasible variables to measure. An instance of this interaction was observed in P9's chat who wanted to explore the relationship between sleep and muscle recovery. Figure 8 shows how **CASE**bot first scoped the goal and collected key context: current sleep habits (which is a candidate independent variable) and how recovery/performance are already measured (candidate dependent variables). It then elicited more detail on workout tracking and feasibility of a consistent sleep routine to understand daily habits and constraints. Finally, it proposed the variables explicitly with specificity -IV: sleep schedule (consistent vs. inconsistent) and DVs: weights/reps, RPE, soreness, and energy.

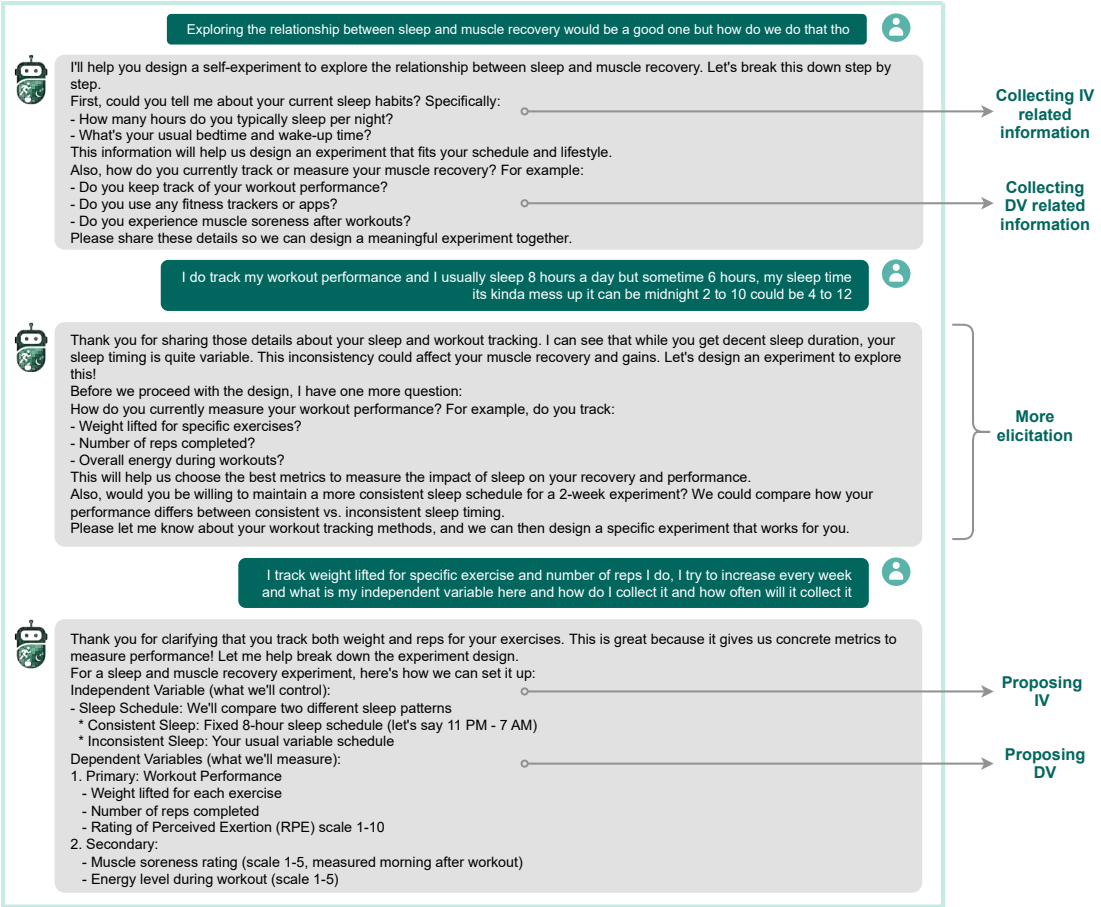

Figure 8: Snapshots of P9's chat transcript: CASEbot collects P9's current context and proposes suitable variables.

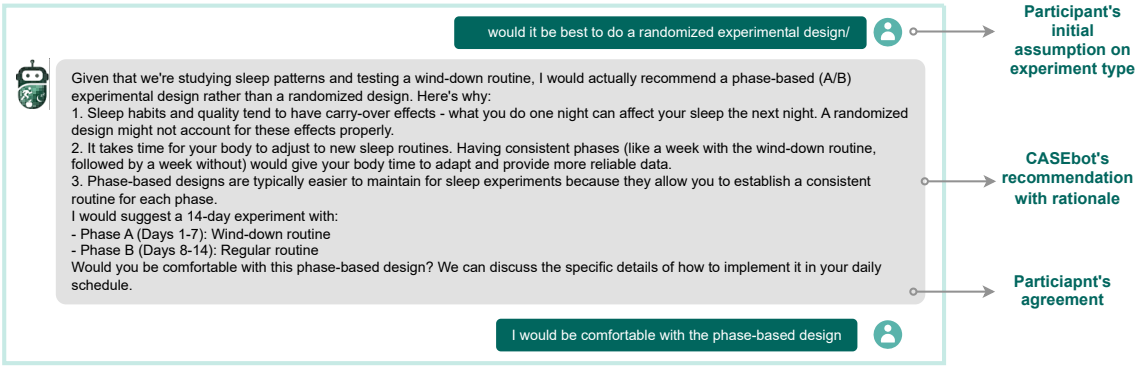

Figure 9: Snapshot of P25's chat transcript: CASEbot recommends the appropriate experiment type

5.2.3 ***Experiment Type***. Overall, **CASE**bot-supported plans scored higher on the Experiment Type criteria than worksheet plans (Figure 6; Z=3.650, p<0.001 and 95% CI 0.232-0.768 points out of 6). Table 4 shows analyses on each 4 sub-criteria of Experiment Type — **4(a)** *Temporal Visibility*, **4(b)** *Appropriate*, **4(c)** *Safety–IV*, and **4(d)** *Safety–DV* along with the aggregate result (see Table 4.3). Plans guided by **CASE**bot achieved higher scores than the worksheet-guided ones on the sub-criterion **4(c)** (Z=3.788, p<0.001, 95% CI=0.161-0.506 points higher out of 2). **CASE**bot plans further scored marginally higher on **4(a)** (Z=1.704, p=0.088, 95% CI -0.014

lower - 0.205 higher out of 1 point). No significant difference was observed between **CASE**bot and worksheet in **4(b)** (p = 0.648) or **4(d)** (p = 0.308). As per Figure 5, participants perceived **CASE**bot to be more effective in *Providing a good experiment design (type)* (Z=5.369, p<0.001 and 95% CI 0.877-1.885 points out of 7).

Table 4: Summary of Results for Experiment Type Criteria: CASEbot-guided plans scored higher as accumulated score (Total) and the 4(c) sub-criteria with statistical significance.

Criteria	Pltfm: CASEbot [95% CI]	z-value	p-value
4 Experiment Type Aggregate	0.500 [0.232, 0.768]	3.650	< 0.001
4(a) Temporal Visibility	0.095 [-0.014, 0.205]	1.704	0.088
4(b) Appropriate	0.024 [-0.078, 0.126]	0.456	0.648
4(c) Safety–IV*	0.333 [0.161, 0.506]	3.788	< 0.001
4(d) Safety–DV*	0.048 [-0.044, 0.139]	1.019	0.308

**Note:* Experiment Type criteria is on a 6-point scale. All sub-criteria are binary (0 or 1) except for 4(c) and 4(d) which are on a 2-point scale (0, 1, or 2)

Participants described that **CASE**bot's approach of interactively investigating aspects of their life resulted in suggesting appropriate and reasonable experiment design type. As P20 said, "***CASE**bot was way better as it asked you very specific questions and considered all factors. It was also great because it automatically generated the design after a series of questions*". P49 appreciated the suggestions **CASE**bot made: "*It [**CASE**bot] filled in the gaps of my knowledge on experiment design, and was very, very accurate and helpful in its suggestions.*" Similarly, P13 commented that they found the **CASE**bot-supported experiment design would help them achieve their self-understanding goal, "***CASE**bot made me answer much more questions and was very useful in finding a reasonable experiment that felt helpful to me. The worksheet it was much much harder to do so.*".

Analyzing the transcripts for the *Experiment Type* phase of the conversation, we noticed **CASE**bot was able to nudge participants to an appropriate type, sometimes even correcting the participant. We observed this example in P25's transcript. As figure 9 shows, P25 had initially chosen randomized experiment type but **CASE**bot instead proposed a phase-based design with a detailed rationale for the choice. P25 ultimately decided to follow **CASE**bot's recommendation.

5.2.4 ***Experiment Schedule***. The accumulated score on Experiment Schedule was higher for **CASE**bot than worksheet (Figure 6; Z=5.891, p<0.001 and 95% CI 0.620-1.238 points higher out of 4). Table 5 shows analyses of Experiment Schedule sub-criteria: (**5(a)** *Sch: Weekly schedule*, **5(b)** *Sch: Starting date*, **5(c)** *Sch: Control days mentioned*, and **5(d)** *Sch: Intervention days mentioned* and the aggregate result (see Table 4.3). **CASE**bot-supported experiments scored higher on the sub-criterion **5(a)** (Z=3.815, p<0.001, 95% CI 0.127-0.396 out of 1 point) and **5(b)** (Z=8.162, p<0.001, 95% CI 0.470-0.768 out of 1 point), suggesting that **CASE**bot plans had a more structured and organized schedule. This result is perhaps a consequence of **CASE**bot's prompts putting heavy emphasis on mentioning the starting day in the schedule and devising a weekly plan. No significant difference was observed across **5(c)** (p = 0.648) and **5(d)** (p = 0.648). As per Figure 5, participants perceived **CASE**bot to be more effective in *Planning the self-experiment schedule* (Z=4.945, p<0.001 and 95% CI 0.762-1.762 points out of 7).

Table 5: Summary of Results for Experiment Schedule Criteria (Experiment Schedule criteria is on a 4-point scale and all sub-criteria are binary (0 or 1)): Except for 5(c) and 5(d), CASEbot outperforms worksheet in rest of them, with significant differences.

Criteria	Pltfm: CASEbot [95% CI]	z-value	p-value
5 Experiment Schedule Aggregate	0.929 [0.620, 1.238]	5.891	< 0.001
5(a) Sch: Weekly schedule	0.262 [0.127, 0.396]	3.815	< 0.001
5(b) Sch: Starting date	0.619 [0.470, 0.768]	8.162	< 0.001
5(c) Sch: Control days mentioned	0.024 [-0.078, 0.126]	0.456	0.648
5(d) Sch: Intervn. days mentioned	0.024 [-0.078, 0.126]	0.456	0.648

Six participants (P3, P6, P29, P32, P38, P46) specifically highlighted the benefits of **CASE**bot's support in the *Experiment Scheduling* phase, while three participants (P16, P37, P51) pointed to advantages of the worksheet. P19 said, "*CASEbot gave a more structured view, where I could ask for a schedule to be shown and sent to me, Worksheet gave a more free hands-off experience though*". P35 opined about the potential difficulty a *novice* might face when using the worksheet: "*over a two week schedule what is the schedule would be unhelpful for someone inexperienced with experiment design, typically both control and intervention periods are of equal length to make them comparable, but a novice may not know that and may just take one day as their control.*" P3, who had some skepticism about **CASE**bot's overall level of guidance, felt that its scheduling support was valuable "***CASE**bot provided lots of guidance, which personally felt like a bit too much. It did, however, generate a very useful schedule for me.*". Participants found that **CASE**bot guided them towards building feasible experiment schedules, while allowing room for adjustment and negotiation: P48 said, "*I gave the CASEBot different intervals of time to eat, and it generated another schedule for me, which was more realistic for my goals and went better with my schedule. Initially, it did convince me that its schedule was better but after discussion we came to a conclusion to go with mine.*".

Analyzing the transcripts, we saw how **CASE**bot is able to collect the relevant information to guide the creation of a well-defined self-experiment plan. Figure 10 shows part of P19's chat transcript focusing on how **CASE**bot guided the participant through identifying broad health goals to specifying areas of interest, current habits, and exercise preferences. Based on the responses, it provided a tailored body-weight routine, and generated a complete 14-day schedule that aligned with the P19's daily routine.

Despite **CASE**bot supporting users in designing a well-structured self-experiment plan, some participants felt the system's guidance as overly controlling, noting that the large number of questions sometimes caused confusion. As P3 reflected regarding **CASE**bot, "*It felt like i was dragged through the process, given too much guidance, and asked redundant questions. In the end I how have an effectively designed experiment, but it is not one I care to participate in*". P9 also reflected, "*It was pretty good, but it asked too many questions at once. It asked multiple questions, and I didn't know which one to answer or if I should answer all of them.*"

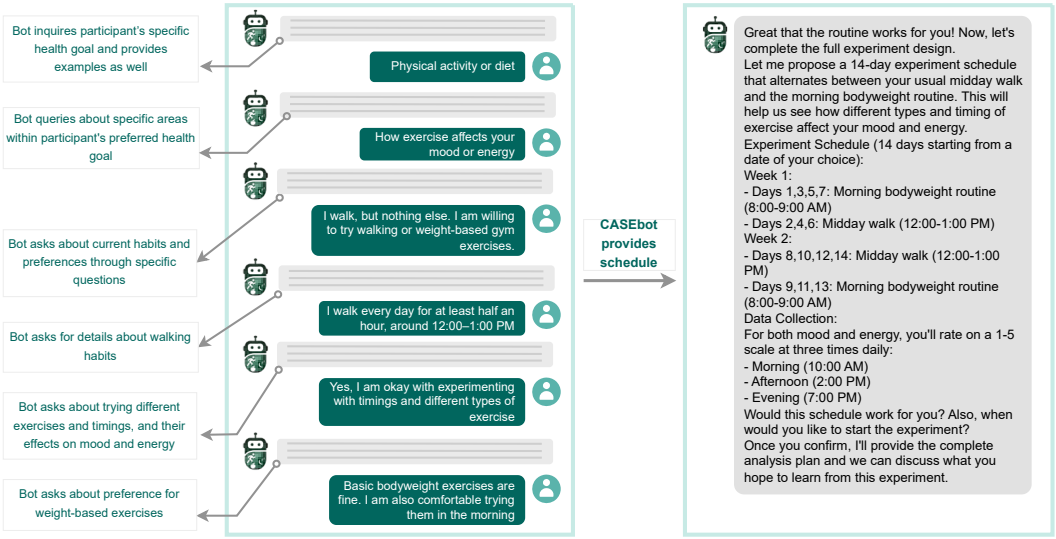

Figure 10: A demonstration of how CASEbot elicits participant's (P19) information and incorporates them into building an experiment schedule.

5.3 Supporting Personalization

Overall, participants found worksheet supported personalization through its flexibility which encouraged autonomy and reflection. On the other hand, participants perceived **CASE**bot to support personalization through interactive questioning; adapting to user goals, habits, and constraints; and the generation of tailored recommendations. Analyzing participants' perception of the personalization provided by the platforms, we found that overall, **CASE**bot-supported designs were deemed to be more personalized (Z=3.435, p=0.001, 95% CI 0.348-1.271 on a 7-point scale).

Given the limited guidance offered by the worksheet, participants often found it to be more flexible and thereby easier to personalize to their needs. As P3 mentioned, "*Because the worksheet provided less guidance, It was much easier to design an experiment that was tailored to my own goals*". On the other hand, participants appreciated the detailed support **CASE**bot's conversational approach provided throughout the process. P11 stated, "*It was very easy to design a self-experiment with **CASE**bot, you just have to have a goal in mind and the rest will just have to be answering the questions from **CASE**bot to create a fitted experiment with your personal habbits or information*". Diving further into how **CASE**bot supported such personalization, we observed two high-level interaction patterns: proactive and reactive.

5.3.1 ***Proactive Personalization***. Participants found that **CASE**bot was able to proactively steer the design process by asking relevant questions to collect contextual information or providing suggestions for how to improve the experiment design. P12's reflection highlighted this proactiveness, "*it asks about my health goals, but it also goes deeper asking specifics about my sleep schedule like what affects it and how I sleep. And then it latches onto a good independent variable and develops the self-experiment from there*". This proactiveness was visible across the different stages of the design process. For example, P29 found the proactive questioning helpful in taking a step back to consider alternatives for variables. "*It was interesting to see the recommendations [for variables] and I think it did a good at making me consider something that I haven't thought of (like flavoring the water)*". P33 similarly appreciated the variable alternatives suggested by **CASE**bot, "*I could not consider points such as wind-down activity or come up with such robust variables without the help of **CASE**bot*". P20 commented that **CASE**bot proactively offered suggestions around timing their meditation activities, saying, "*It would ask me what time in the day I can do the meditation regularly and what times in the day I feel I can consistently record the anxiety data. I wouldn't have given the timings of these things as much thought if I designed it on my own*".

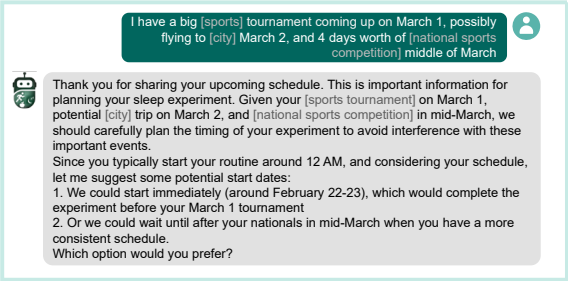

Figure 11: Excerpts of P25's scheduling constraints and CASEbot's response demosntrating its reactive personalization capability.

5.3.2 ***Reactive Personalization***. In addition to offering suggestions, participants appreciated **CASE**bot's ability to reactively be receptive of the constraints and requirements

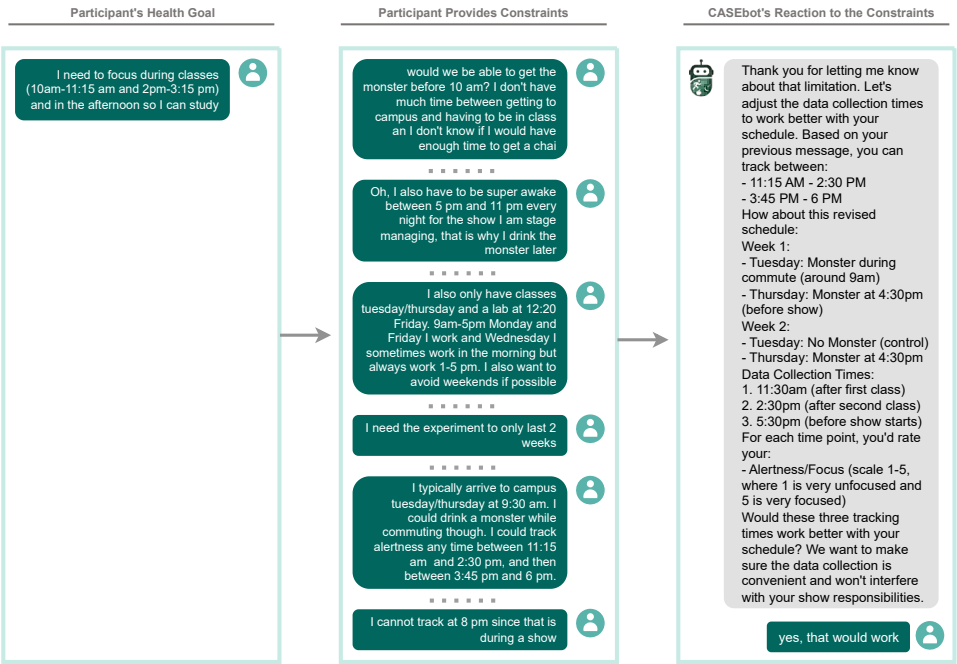

Figure 12: Excerpts of P16's chat transcript reflecting on CASEbot's reactive personalization capability: CASEbot incorporated all the constraints.

surfaced by the participants. P25 disclosed their upcoming travel (obfuscated for anonymity) constraints and **CASE**bot took them into consideration and recommended two alternatives as depicted in Figure 11.

P8 appreciated **CASE**bot's ability to not only provide guidance, but to improve upon their experiment suggestions, "*It provides clear guidance on variables, analysis plans, timelines,... and even improves on my suggestions which are much more helpful that just a list of questions in the checklist*". In a similar vein, P16 had several scheduling constraints that they were able to provide to **CASE**bot that were successfully integrated in the updated design (see Figure 12).

5.4 Safety Assurance

Overall, the worksheet implicitly guided participants towards safer experiments by prompting them to be mindful of experiment parameters. On the other hand, **CASE**bot incorporated safety through guiding participants away from potentially unsafe parameter choices, recommending gradual changes from current baseline instead of drastic ones, and assessed feasibility of participants realistically undertaking the experiment. On the quantitative front, as mentioned earlier (section 5.2.3), plans guided by **CASE**bot achieved higher scores than the worksheet-guided ones on **4(c)** *Safety: IV* (Z=3.788, p<0.001, 95% CI 0.161-0.506 points higher out of 2). No significant difference was observed for **4(d)** *Safety: DV* (p = 0.308). Figure 13 shows the scores for both platform. Participants' perception on safety for **CASE**bot was

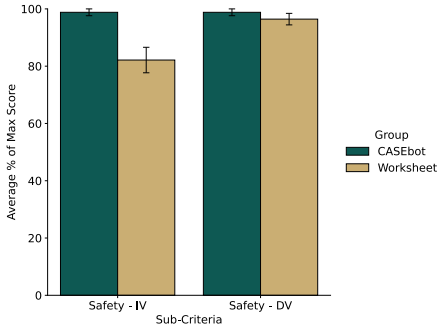

Figure 13: Percentage of maximum score for the safety related sub-criteria: In both IV and DV related safety.

marginally higher (Z=1.704, p=0.088, 95% CI -0.014 lower - 0.205 higher out of 1 point).

Analyzing the chat transcripts, we discovered multiple cases where **CASE**bot raised safety concerns during the self-experimentation planning. For example, when P6 proposed an experiment to drink more water – moving from 1.8-2.4 liters a day to 4 liters a day – **CASE**bot interjected suggesting a more gradual increase. "*Four liters (about 135 oz) is quite a significant increase from your current intake of 1.8-2.4 liters. Let's design a gradual increase to make it more sustainable and safer*". **CASE**bot was proactive about highlighting the safety limit for awareness, while

confirming that the proposed experiment was within the safe limits. For example, when P16 decided to test the effect of the time of caffeine consumption on their focus, **CASE**bot responded with: "*Note: I want to ensure we design an experiment that's both safe and effective. The FDA suggests limiting caffeine intake to no more than 400mg per day for healthy adults.*"

We also observed a situation in which although the experiment would ideally be safe, due to the physical injury of the individual, it asked the participant to instead consult with their provider. P34 wanted to vary the frequency of their physical therapy to reduce pains/swelling which were happening due to ACL and UCL sprains. **CASE**bot reiterated the safety criteria of experiments, and advised on sticking to the guidelines provided by the participant's doctor: "*we need to ensure any self-experiment we design is safe and doesn't interfere with your recovery process. Also, while we can explore ways to manage pain and swelling, please note that we should focus on activities that your doctor has approved*". Overall, **CASE**bot first assessed the risk factors associated with the intended experiment, and put emphasis on the safety concern by providing medical disclaimer or FDA approved guidelines.

We also saw a scenario where **CASE**bot redirected the individual given the experiment would be outside of its scope – not pertaining to physical activity, diet, or sleep. P35 wanted to test the effects of Ashwaganda on anxiety, and **CASE**bot responded with: "*I need to redirect this goal since supplements like ashwagandha fall outside the scope of safe self-experimentation that I can help with. Instead, let's focus on lifestyle factors that might help reduce your jitters/leg shaking*.". However, we also observed cases (e.g., P20) where **CASE**bot-supported plans that are beyond the specified domains (physical activity, diet, or sleep) (e.g., observing effect of meditation on anxiety).

Concerningly, we also saw a few instances where **CASE**bot presented people with incorrect information. When discussing about water intake with P29, it responded with "*Two cups of water per day is quite low - the general recommendation for adults is about 8 cups (64 ounces) per day, though this can vary based on factors like activity level and climate.*" According to the Mayo clinic [68], this is close to half the recommended amount (15.5 cups for men and 11.5 cups for women).

.

Table 6 synthesizes the findings across the dimensions: Structure, Personalization and Safety.

6 Discussion

We designed **CASE**bot with the goal of making the practice of self-experimentation more accessible by scaffolding the expertise required in the design of such experiment. Our evaluation shows that the conversational interactivity provided by **CASE**bot supported the participants in designing well-structured, personalized, and safe experiments as compared to a static worksheet. Although successful, we also see room for future improvement in supporting the practice.

6.1 Tension Between Guidance vs. Autonomy

Although people found the support helpful in general, our findings surfaced some hidden tensions surrounding the interactivity, primarily, the tension between providing guidance vs. supporting autonomy. We observed two clear instances of this trade-off being surfaced during our analysis. For instance, **CASE**bot's detailed, probing questions were seen by many as instrumental in creating a robust and feasible plan that considered factors they would have otherwise missed (P20, P29, and P33 in Section 5.3.1). However, as described in Section 5.2, we also saw participants who found the same interaction to be a source of friction, describing it as having too many questions and feeling overwhelmed (P9), ultimately slowing them down and making the process feel inefficient. A similar tension was also observed surrounding control over the experiment's direction. Many participants valued **CASE**bot's active role in steering the design, appreciating its clear guidance on variables, timelines, and analysis, and even its ability to improve upon their initial suggestions. While some participants found this "guidance to be too strong" and steering them away from their plans. This may well be a difference between people who have clear idea of what experiment they want to design, and want to be done with it as soon as possible vs. people who are open to input and want to spend the time to design their ideal experiment. Moreover, our findings should be interpreted in light of the relatively high digital literacy of the participant pool, allowing them more autonomy in engaging with the system. For broader population with low digital and statistical literacy, the same interaction may require stronger guidance.

This tension indicates a need for better control over interactivity by adjusting guidance levels while preserving user autonomy. Similar tensions emerged in prior work, such as TummyTrials, where some participants preferred custom symptom scales while others wanted more flexible experiment design [47]. Future systems could dynamically tailor the depth and pacing of support – echoeing Galileo [76] that employs just-in-time scaffolding and lets users navigate experiment components on their own terms. In our context, early assessments of user expertise could help provide detailed feedback for novices and concise, targeted feedback for experienced users [1, 31]. Alternatively, user could explicitly control when and how to get guidance, such as drafting an experiment independently and requesting feedback afterward for potential issues and specifics, or adjusting the verbosity filter to set how *chatty* the system should be [43]. Collectively, these strategies highlight the value of adaptive support rather than a simple choice between highly directive and fully hands-off approaches.

6.2 Balancing Access with Rigor and Safety

Because we understand the experiment design process to a degree where we can break it down into a step-by-step process, it helped us leverage LLMs fill the gaps around the range of experiments people would be interested in conducting while working around their individual constraints. However, the notion of *control* is a tenuous one when it comes to LLMs. We are dealing with a double-edged sword where on one hand it allows us to navigate ill-defined challenges but on the other we have limited control over it and it can behave unpredictably.

Similar to prior research which surfaced a trade-off between scientific rigor and flexibility [47, 48, 98], we noticed an emerging tension between the desire to maintain high scientific rigor and

Table 6: Summary of findings across Structure, Personalization, and Safety dimensions

	CASEbot's Strengths	CASEbot's Gaps	Design Recommendations
Structure	• **CASE**bot-guided plans scored higher across all structural criteria, with significant improvement in scheduling, IV specificity (dosage/timing), and DV measurability. • Easy to use. • Provides well-structured guidance through clear, reasonable questions. • Helps participants turn vague goals into specific hypotheses.	• Some participants felt the guidance was too strong. • Asking multiple questions in the same message caused confusion.	• Adapt guidance to user expertise – more scaffolding for beginners, streamlined prompts for advanced users. • More user control over interaction style, including response verbosity.
Personalization	• **CASE**bot proactively seeks detailed user contexts to generate personalized recommendations. • **CASE**bot successfully and promptly incorporated user-provided constraints into experiment plans.	• Some participants preferred worksheet over **CASE**bot as it afforded more freedom and personalization scope.	• Let users control when they receive feedback, choosing between continuous or on-demand guidance.
Safety	• Most **CASE**bot-guided plans met safety requirements, reflected in their high safety related sub-criteria scores. • **CASE**bot proactively screens unsafe variable conditions and recommends safer alternatives.	• Some **CASE**bot-guided plans extended beyond the specified domains (physical activity, diet, sleep), raising safety concerns.	• Incorporate expert-in-the-loop through multi-agent mechanisms where agents review the designs, make summary for expert's validation, or query the designs interactively.

the desire to support a wide range of experiments. During the variable selection stage of the conversation, **CASE**bot worked with participants to define their own measures for collecting the data. Often these would be standard 5-point scales for focus, energy level, etc. but other times it would be slightly difficult to measure like *a 45-minute wind down routine* (as P33 mentioned in Section 5.3.1) or *30-minutes of stretching* (P27). Such loosely specified measures introduce potential for compromising internal validity of the experiment, echoing concerns raised in prior work [25, 47].

We also noticed that although **CASE**bot was good at following the prompt instructions around limiting the experiments to physical activity, diet, and sleep, there were instances where it ignored those instructions. For example, as we described in Section 5.4, when P35 wanted to experiment on the effects of *ashwagandha*, **CASE**bot successfully pivoted it to a different design. On the other hand, it supported P20 in designing an experiment to observe the effect of meditation on their anxiety (see Section 5.4). This shows the need for more stringent guardrails and guarantee on its enforcement.

We see two potential paths forward – designing better human-AI collaboration and improving the LLM designs. On the human-AI collaboration front, in the near-term, we see involving experts as the quickest way to navigate the above challenges. Similar to Galileo [76], we can build designs that support experts in quickly reviewing designs to increase their throughput such as leveraging multi-agent systems to evaluate the experiments and provide summaries for the expert to review or to query the designs interactively. On the genAI front, retrieval-augmented generation (RAG) has been proposed to navigate some issues around providing better instructions [59, 81]. However, RAGs require a training dataset [64], which does not exist for self-experiments. It may be a worthwhile exploration to create a synthetic dataset using the known constraints and using that to train the models. Recent research has shown that using a group of LLMs to analyze and interpret complex health and wellbeing data performs better than existing RAG models [18].

6.3 A Vision for GenAI in Self-Experimentation

In our work, we demonstrated that **CASE**bot was able to target a key challenge in the design stage of self-experiments – scaffolding the scientific expertise needed in designing valid experiment while providing personalization and safety. Recent work demonstrates the potential of genAI in supporting all stages of self-tracking, including deciding what data are relevant, determining appropriate strategies for collecting them, adjusting tracking plans, and making sense of insights those data produce [17]. Building on these opportunities, we posit that genAI can also be leveraged to tackle known challenges in the other stages: execution, analysis, reflection, and next steps [16, 25, 47].

A core challenge surrounding execution step is data quality – did people actually follow the experiment instructions or did they only

think they followed them [47]. Based on recent literature, we are seeing examples where genAI is being used to enhance the process of preparation [86] and reflection by enabling people to engage in a query-based dialogue to explore and understand their data [45]. Leveraging genAI, we can imagine developing designs that not only help people collect the data, but also help with collecting additional information pertaining to the quality of the data. One can imagine adopting a learning-based approach where the system can ask the person a multiple choice question as part of their daily data collection that tests their mental model of the experiment.

Once the data is collected, the individual needs to analyze and reflect on the process. Unlike standard self-tracking, where analysis is typically temporal aggregation (e.g., average steps per day/week/month), the analysis in self-experimentation involves testing a hypothesis. Most lay people do not have sufficient scientific training to know what *testing hypothesis* means or how to interpret *standard* statistical measures like p-values. Past research has surfaced that even when people are presented with visualizations and plain language interpretations of statistical analyses, different people still interpret the same results differently. Assuming that the system has access to the experiment design and collected data, it can work with the user to figure out their preferred method of looking at the findings and walk through the results in an interactive manner to improve comprehension [69].

Finally, the question of what to do after an experiment is concluded is an open challenge. The experiment is not an end in itself but, a part of an individual's larger health journey. Past research has primarily focused on the experimentation process and has been unable to provide much structured support to individuals on what to do once the experiment is concluded. Primarily, such research has proposed following up with alternate experiments to run (e.g., testing a different hypothesis or repeating the experiment with a longer duration for more confidence) [25, 47]. In our design, we intentionally situate the self-experiment within an individuals broader health goal to better support such post-experiment pathways. We can imagine leveraging genAI to contextualize the findings of the experiment within the goal and presenting the individual with personalized options on what to do next. For example, if the experiment shows that a 30-minute walk in the evening leads to improved sleep quality, it can work with the individual to explore how they can integrate the new behavior longer-term, or explore alternate behaviors that maybe easier to sustain and test them through experimentation.

Overall, we see genAI having the potential to meaningful impact on making the self-experimentation practice more accessible. We do not think of genAI as a *silver bullet* that will solve everything but rather as a new tool for us to use to navigate problem spaces that are ill-defined and lack concrete specification.

7 Conclusion

With **CASE**bot, we demonstrated how a genAI-enabled conversational agent is able to combine structured instructions with contextual and preferential information to interactively scaffold the design of self-experiments, thereby enabling people to create personalized experiments that adhere to standards of experimental rigor and safety. In comparison to a static worksheet, participants were able to design better experiments according to an objective rubric and in their subjective opinions. Our analysis also revealed tensions around the amount of guidance provided versus the autonomy afforded in the process, pointing to a need for better user-control on the interactivity in future explorations. Finally, we contribute our cautiously optimistic vision for the future of genAI in making the broader practice of self-experimentation more accessible, rigorous, safe, and empowering.

Acknowledgments

We would like to thank our colleagues who helped us test our system prior to data collection, ensuring a smooth study; the participants who consented to share their data for use in this research; and the anonymous reviewers for their constructive feedback. We also thank the CHI Bootcamp at UCI for their feedback on drafts of the paper. This work was funded in part by the National Science Foundation under awards IIS-2237389 and IIS-2441333.

References

[1] Sungeun An, Robert Moore, Eric Young Liu, and Guang-Jie Ren. 2021. Recipient design for conversational agents: Tailoring agent's utterance to user's knowledge. In *Proceedings of the 3rd Conference on Conversational User Interfaces*. 1–5.

[2] Anthropic. 2025. Claude. Large language model. URL: https://claude.ai.

[3] John W Ayers, Adam Poliak, Mark Dredze, Eric C Leas, Zechariah Zhu, Jessica B Kelley, Dennis J Faix, Aaron M Goodman, Christopher A Longhurst, Michael Hogarth, et al. 2023. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. *JAMA internal medicine* 183, 6 (2023), 589–596.

[4] Amid Ayobi, Paul Marshall, and A Cox. 2017. Self-Experimentation and the Value of Uncertainty. ACM.

[5] Amid Ayobi, Tobias Sonne, Paul Marshall, and Anna L. Cox. 2018. Flexible and Mindful Self-Tracking: Design Implications from Paper Bullet Journals. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3173574.3173602

[6] Colin Barr, Maria Marois, Ida Sim, Christopher H Schmid, Barth Wilsey, Deborah Ward, Naihua Duan, Ron D Hays, Joshua Selsky, Joseph Servadio, et al. 2015. The PREEMPT study-evaluating smartphone-assisted n-of-1 trials in patients with chronic pain: study protocol for a randomized controlled trial. *Trials* 16, 1 (2015), 67.

[7] Vince Bartle, Janice Lyu, Freesoul El Shabazz-Thompson, Yunmin Oh, Angela Anqi Chen, Yu-Jan Chang, Kenneth Holstein, and Nicola Dell. 2022. "a second voice": Investigating opportunities and challenges for interactive voice assistants to support home health aides. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17.

[8] Marit Bentvelzen, Jasmin Niess, Mikołaj P Woźniak, and Paweł W Woźniak. 2021. The development and validation of the technology-supported reflection inventory. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–8.

[9] Marit Bentvelzen, Paweł W Woźniak, Pia SF Herbes, Evropi Stefanidi, and Jasmin Niess. 2022. Revisiting reflection in hci: Four design resources for technologies that support reflection. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 1 (2022), 1–27.

[10] Tom Bocklisch, Joey Faulkner, Nick Pawlowski, and Alan Nichol. 2017. Rasa: Open source language understanding and dialogue management. *arXiv preprint arXiv:1712.05181* (2017).

[11] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.

[12] Isis Bulté and Patrick Onghena. 2011. When the truth hits you between the eyes. *Methodology* (2011).

[13] David Byrne. 2022. A worked example of Braun and Clarke's approach to reflexive thematic analysis. *Quality & quantity* 56, 3 (2022), 1391–1412.

[14] Centers for Disease Control and Prevention. 2024. Nutrition Guidelines and Recommendations. https://www.cdc.gov/nutrition/php/guidelines-recommendations/index.html.

[15] Eun Kyoung Choe, Saeed Abdullah, Mashfiqui Rabbi, Edison Thomaz, Daniel A Epstein, Felicia Cordeiro, Matthew Kay, Gregory D Abowd, Tanzeem Choudhury,

James Fogarty, et al. 2017. Semi-automated tracking: a balanced approach for self-monitoring applications. *IEEE Pervasive Computing* 16, 1 (2017), 74–84.

[16] Eun Kyoung Choe, Nicole B Lee, Bongshin Lee, Wanda Pratt, and Julie A Kientz. 2014. Understanding quantified-selfers' practices in collecting and exploring personal data. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1143–1152.

[17] Shaan Chopra, Katherine Juarez, James Fogarty, and Sean A Munson. 2025. Engagements with Generative AI and Personal Health Informatics: Opportunities for Planning, Tracking, Reflecting, and Acting around Personal Health Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 9, 3 (2025), 1–33.

[18] Akshat Choube, Ha Le, Jiachen Li, Kaixin Ji, Vedant Das Swain, and Varun Mishra. 2025. GLOSS: Group of LLMs for Open-Ended Sensemaking of Passive Sensing Data for Health and Wellbeing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 9, 3 (2025), 1–32.

[19] Kevin Corti, Geetha Reddy, Ellen Choi, and Alex Gillespie. 2015. The researcher as experimental subject: Using self-experimentation to access experiences, understand social phenomena, and stimulate reflexivity. *Integrative Psychological and Behavioral Science* 49, 2 (2015), 288–308.

[20] Aykut Coşkun and Armağan Karahanoğlu. 2023. Data sensemaking in self-tracking: Towards a new generation of self-tracking tools. *International Journal of Human–Computer Interaction* 39, 12 (2023), 2339–2360.

[21] Jesse Dallery, Rachel N Cassidy, and Bethany R Raiff. 2013. Single-case experimental designs to evaluate novel technology-based health interventions. *Journal of medical Internet research* 15, 2 (2013), e22.

[22] Nediyana Daskalova, Karthik Desingh, Alexandra Papoutsaki, Diane Schulze, Han Sha, and Jeff Huang. 2017. Lessons learned from two cohorts of personal informatics self-experiments. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 1, 3 (2017), 1–22.

[23] Nediyana Daskalova, Eindra Kyi, Kevin Ouyang, Arthur Borem, Sally Chen, Sung Hyun Park, Nicole Nugent, and Jeff Huang. 2021. Self-e: Smartphone-supported guidance for customizable self-experimentation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–13.

[24] Nediyana Daskalova, Danaë Metaxa-Kakavouli, Adrienne Tran, Nicole Nugent, Julie Boergers, John McGeary, and Jeff Huang. 2016. SleepCoacher: A personalized automated self-experimentation system for sleep recommendations. In *Proceedings of the 29th annual symposium on user interface software and technology*. 347–358.

[25] Nediyana Daskalova, Jina Yoon, Yibing Wang, Cintia Araujo, Guillermo Beltran Jr, Nicole Nugent, John McGeary, Joseph Jay Williams, and Jeff Huang. 2020. SleepBandits: Guided flexible self-experiments for sleep. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.

[26] C David and J Paul. 2023. ChatGPT and large language models: what's the risk.

[27] Eric J Daza and Logan Schneider. 2022. Model-twin randomization (motr): A Monte Carlo method for estimating the within-individual average treatment effect using wearable sensors. *arXiv preprint arXiv:2208.00739* (2022).

[28] George T Doran. 1981. There's a SMART way to write managements's goals and objectives. *Management review* 70, 11 (1981).

[29] Adam G Dunn, Ivy Shih, Julie Ayre, and Heiko Spallek. 2023. What generative AI means for trust in health communications. *Journal of Communication in Healthcare* 16, 4 (2023), 385–388.

[30] Eugene Edgington and Patrick Onghena. 2007. *Randomization tests*. Chapman and Hall/CRC.

[31] Joy Egede, Maria J Galvez Trigo, Adrian Hazzard, Martin Porcheron, Edgar Bodiaj, Joel E Fischer, Chris Greenhalgh, and Michel Valstar. 2021. Designing an adaptive embodied conversational agent for health literacy: a user study. In *Proceedings of the 21st ACM International Conference on Intelligent Virtual Agents*. 112–119.

[32] Tina Ekhtiar, Armağan Karahanoğlu, Rúben Gouveia, and Geke Ludden. 2023. Goals for goal setting: a scoping review on personal informatics. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 2625–2641.

[33] Zachary Englhardt, Chengqian Ma, Margaret E Morris, Chun-Cheng Chang, Xuhai" Orson" Xu, Lianhui Qin, Daniel McDuff, Xin Liu, Shwetak Patel, and Vikram Iyer. 2024. From classification to clinical insights: Towards analyzing and reasoning about mobile and behavioral health data with large language models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 2 (2024), 1–25.

[34] Daniel A Epstein, An Ping, James Fogarty, and Sean A Munson. 2015. A lived informatics model of personal informatics. In *Proceedings of the 2015 ACM international joint conference on pervasive and ubiquitous computing*. 731–742.

[35] Shan Feng, Matti Mäntymäki, Amandeep Dhir, and Hannu Salmela. 2021. How self-tracking and the quantified self promote health and well-being: systematic review. *Journal of Medical Internet Research* 23, 9 (2021), e25171.

[36] Markus A Feufel and S Frederica Stahl. 2012. What do web-use skill differences imply for online health information searches? *Journal of medical Internet research* 14, 3 (2012), e87.

[37] Brian J Fogg. 2020. *Tiny habits: The small changes that change everything*. Harvest.

[38] Silvia Fornasini et al. 2021. Clinical self-tracking to improve knowing in practice: designing self-experiments for Type 2 Diabetes care. (2021).

[39] Elliot G. Mitchell, Elizabeth M. Heitkemper, Marissa Burgermaster, Matthew E. Levine, Yishen Miao, Maria L. Hwang, Pooja M. Desai, Andrea Cassells, Jonathan N. Tobin, Esteban G. Tabak, et al. 2021. From reflection to action: Combining machine learning with expert knowledge for nutrition goal recommendations. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–17.

[40] Asghar Ghasemi, Farhad Hosseinpanah, Khosrow Kashfi, and Zahra Bahadoran. 2025. Research hypothesis: a brief history, central role in scientific inquiry, and characteristics. *Addiction & Health* 17 (2025), 1623.

[41] Per M Hellström. 2006. This year's Nobel Prize to gastroenterology: Robin Warren and Barry Marshall awarded for their discovery of Helicobacter pylori as pathogen in the gastrointestinal tract. *World Journal of Gastroenterology: WJG* 12, 19 (2006), 3126.

[42] Mieke Heyvaert and Patrick Onghena. 2014. Randomization tests for single-case experiments: State of the art, state of the science, and state of the application. *Journal of Contextual Behavioral Science* 3, 1 (2014), 51–64.

[43] Shih-Hong Huang, Ya-Fang Lin, Zeyu He, Chieh-Yang Huang, and Ting-Hao Kenneth Huang. 2024. How does conversation length impact user's satisfaction? A case study of length-controlled conversations with LLM-powered chatbots. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–13.

[44] Rune Møberg Jacobsen, Samuel Rhys Cox, Carla F Griggio, and Niels Van Berkel. 2025. Chatbots for Data Collection in Surveys: A Comparison of Four Theory-Based Interview Probes. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–21.

[45] Matthew Jörke, Shardul Sapkota, Lyndsea Warkenthien, Niklas Vainio, Paul Schmiedmayer, Emma Brunskill, and James A Landay. 2025. GPTCoach: Towards LLM-Based Physical Activity Coaching. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–46.

[46] Ravi Karkar, James Fogarty, Julie A Kientz, Sean A Munson, Roger Vilardaga, and Jasmine Zia. 2015. Opportunities and challenges for self-experimentation in self-tracking. In *Adjunct Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Computers*. 991–996.

[47] Ravi Karkar, Jessica Schroeder, Daniel A Epstein, Laura R Pina, Jeffrey Scofield, James Fogarty, Julie A Kientz, Sean A Munson, Roger Vilardaga, and Jasmine Zia. 2017. Tummytrials: a feasibility study of using self-experimentation to detect individualized food triggers. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 6850–6863.

[48] Ravi Karkar, Jasmine Zia, Roger Vilardaga, Sonali R Mishra, James Fogarty, Sean A Munson, and Julie A Kientz. 2016. A framework for self-experimentation in personalized health. *Journal of the American Medical Informatics Association* 23, 3 (2016), 440–448.

[49] Alan E Kazdin. 2020. *Single-Case Research Designs: Methods for Clinical and Applied Settings*. Oxford University Press;.

[50] Alan E Kazdin. 2021. Single-case experimental designs: Characteristics, changes, and challenges. *Journal of the Experimental Analysis of Behavior* 115, 1 (2021), 56–85.

[51] Alan E Kazdin and A Hussain Tuma. 1982. Single-case research designs. (1982).

[52] Joanna Kempner and John Bailey. 2019. Collective self-experimentation in patient-led research: How online health communities foster innovation. *Social Science & Medicine* 238 (2019), 112366.

[53] Ian Kerridge. 2003. Altruism or reckless curiosity? A brief history of self experimentation in medicine. *Internal medicine journal* 33, 4 (2003).

[54] Elisabeth T Kersten-van Dijk, Joyce HDM Westerink, Femke Beute, and Wijnand A IJsselsteijn. 2017. Personal informatics, self-insight, and behavior change: A critical review of current literature. *Human–Computer Interaction* 32, 5-6 (2017), 268–296.

[55] Taewan Kim, Seolyeong Bae, Hyun Ah Kim, Su-woo Lee, Hwajung Hong, Chanmo Yang, and Young-Ho Kim. 2024. MindfulDiary: Harnessing large language model to support psychiatric patients' journaling. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–20.

[56] Predrag Klasnja, Eric B Hekler, Elizabeth V Korinek, John Harlow, and Sonali R Mishra. 2017. Toward usable evidence: optimizing knowledge accumulation in HCI research on health behavior change. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 3071–3082.

[57] Richard L Kravitz, Naihua Duan, Sunita Vohra, Jiang Li, and DEcIDE Methods Center N of 1 Guidance Panel. 2014. Introduction to N-of-1 trials: indications and barriers. *Design and implementation of N-of-1 trials: a user's guide* (2014), 1–11.

[58] Jisoo Lee, Erin Walker, Winslow Burleson, Matthew Kay, Matthew Buman, and Eric B Hekler. 2017. Self-experimentation for behavior change: Design and formative evaluation of two approaches. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 6837–6849.

[59] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.

[60] Ian Li, Anind Dey, and Jodi Forlizzi. 2010. A stage-based model of personal informatics systems. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 557–566.

[61] Ian Li, Anind K Dey, and Jodi Forlizzi. 2011. Understanding my data, myself: supporting self-reflection with ubicomp technologies. In *Proceedings of the 13th international conference on Ubiquitous computing*. 405–414.

[62] Stella Li, Vidhisha Balachandran, Shangbin Feng, Jonathan Ilgen, Emma Pierson, Pang Wei W Koh, and Yulia Tsvetkov. 2024. Mediq: Question-asking llms and a benchmark for reliable interactive clinical reasoning. *Advances in Neural Information Processing Systems* 37 (2024), 28858–28888.

[63] Elizabeth O Lillie, Bradley Patay, Joel Diamant, Brian Issell, Eric J Topol, and Nicholas J Schork. 2011. The n-of-1 clinical trial: the ultimate strategy for individualizing medicine? *Personalized medicine* 8, 2 (2011), 161–173.

[64] Xinyu Lin, Wenjie Wang, Yongqi Li, Shuo Yang, Fuli Feng, Yinwei Wei, and Tat-Seng Chua. 2024. Data-efficient Fine-tuning for LLM-based Recommendation. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*. 365–374.

[65] Deborah Lupton. 2016. *The quantified self*. John Wiley & Sons.

[66] R. H. Major. 1938. Santorio Santorio. *Annals of Medical History* 10, 5 (sep 1938), 369–381.

[67] Lena Mamykina, Daniel A. Epstein, Predrag Klasnja, Donna Sprujt-Metz, Jochen Meyer, Mary Czerwinski, Tim Althoff, Eun Kyoung Choe, Munmun De Choudhury, and Brian Lim. 2022. Grand challenges for personal informatics and AI. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.

[68] Mayo Clinic Staff. 2022. Water: How much should you drink every day? https://www.mayoclinic.org/healthy-lifestyle/nutrition-and-healthy-eating/in-depth/water/art-20044256.

[69] Mike A Merrill, Akshay Paruchuri, Naghmeh Rezaei, Geza Kovacs, Javier Perez, Yun Liu, Erik Schenck, Nova Hammerquist, Jake Sunshine, Shyam Tailor, et al. 2024. Transforming wearable data into health insights using large language model agents. *arXiv preprint arXiv:2406.06464* (2024).

[70] Sean A Munson, Jessica Schroeder, Ravi Karkar, Julie A Kientz, Chia-Fang Chung, and James Fogarty. 2020. The importance of starting with goals in N-of-1 studies. *Frontiers in digital health* 2 (2020), 3.

[71] Subigya Nepal, Arvind Pillai, William Campbell, Talie Massachi, Michael V Heinz, Ashmita Kunwar, Eunsol Soul Choi, Xuhai Xu, Joanna Kuc, Jeremy F Huckins, et al. 2024. MindScape study: integrating LLM and behavioral sensing for personalized AI-driven journaling experiences. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 8, 4 (2024), 1–44.

[72] Allen Neuringer. 1981. Self-experimentation: A call for change. *Behaviorism* 9, 1 (1981), 79–94.

[73] Kannan Nova. 2023. Generative AI in healthcare: advancements in electronic health records, facilitating medical languages, and personalized patient care. *Journal of Advanced Analytics in Healthcare Management* 7, 1 (2023), 115–131.

[74] Patrick Onghena and Eugene S Edgington. 2005. Customization of pain treatments: Single-case design and analysis. *The Clinical journal of pain* 21, 1 (2005), 56–68.

[75] Vineet Pandey, Amnon Amir, Justine Debelius, Embriette R Hyde, Tomasz Kosciolek, Rob Knight, and Scott Klemmer. 2017. Gut instinct: Creating scientific theories with online learners. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 6825–6836.

[76] Vineet Pandey, Tushar Koul, Chen Yang, Daniel McDonald, Mad Price Ball, Bastian Greshake Tzovaras, Rob Knight, and Scott Klemmer. 2021. Galileo: citizen-led experimentation using a social computing system. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–14.

[77] Jaana Parviainen and Juho Rantala. 2022. Chatbot breakthrough in the 2020s? An ethical reflection on the trend of automated consultations in health care. *Medicine, health care and philosophy* 25, 1 (2022), 61–71.

[78] Seth Roberts. 2004. Self-experimentation as a source of new ideas: Ten examples about sleep, mood, health, and weight. *Behavioral and brain sciences* 27, 2 (2004), 227–262.

[79] Seth Roberts and Allen Neuringer. 1998. Self-experimentation. In *Handbook of research methods in human operant behavior*. Springer, 619–655.

[80] John Rooksby, Mattias Rost, Alistair Morrison, and Matthew Chalmers. 2014. Personal tracking as lived informatics. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 1163–1172.

[81] Alireza Salemi and Hamed Zamani. 2024. Evaluating retrieval quality in retrieval-augmented generation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2395–2400.

[82] Akane Sano, Sara Taylor, Craig Ferguson, Akshay Mohan, and Rosalind W Picard. 2017. Quantifyme: An automated single-case experimental design platform. In *International Conference on Wireless Mobile Communication and Healthcare*. Springer, 199–206.

[83] Jessica Schroeder, Ravi Karkar, Natalia Murinova, James Fogarty, and Sean A Munson. 2019. Examining opportunities for goal-directed self-tracking to support chronic condition management. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 3, 4 (2019), 1–26.

[84] Yasaman S Sefidgar, Carla L Castillo, Shaan Chopra, Liwei Jiang, Tae Jones, Anant Mittal, Hyeyoung Ryu, Jessica Schroeder, Allison Cole, Natalia Murinova, et al. 2024. MigraineTracker: Examining patient experiences with goal-directed self-tracking for a chronic health condition. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–19.

[85] Ashish Sharma, Kevin Rushton, Inna Wanyin Lin, Theresa Nguyen, and Tim Althoff. 2024. Facilitating self-guided mental health interventions through human-language model interaction: A case study of cognitive restructuring. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–29.

[86] Donghoon Shin, Gary Hsieh, and Young-Ho Kim. 2025. PlanFitting: Personalized Exercise Planning with Large Language Model-driven Conversational Agent. In *Proceedings of the 7th ACM Conference on Conversational User Interfaces*. 1–19.

[87] Inhwa Song, SoHyun Park, Sachin R Pendse, Jessica Lee Schleider, Munmun De Choudhury, and Young-Ho Kim. 2025. Exploreself: Fostering user-driven exploration and reflection on personal challenges with adaptive guidance by large language models. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–22.

[88] Konstantin R Strömel, Stanislas Henry, Tim Johansson, Jasmin Niess, and Paweł W Woźniak. 2024. Narrating fitness: Leveraging large language models for reflective fitness tracker data interpretation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–16.

[89] Nancy Sturman, Jane Nikles, Peter Worthy, Shauna Fjaagesund, Rachel Elphinston, Nicole Andrews, Michele Sterling, and Stefan Konigorski. 2024. Evaluating digitally-enabled self-experiments in general practice to support patients managing persistent pain. *medRxiv* (2024), 2024–12.

[90] Tao Tu, Anil Palepu, Mike Schaekermann, Khaled Saab, Jan Freyberg, Ryutaro Tanno, Amy Wang, Brenna Li, Mohamed Amin, Nenad Tomasev, et al. 2024. Towards conversational diagnostic AI. *arXiv preprint arXiv:2401.05654* (2024).

[91] U.S. Food and Drug Administration. 2025. FDA's Nutrition Initiatives. https://www.fda.gov/food/nutrition-food-labeling-and-critical-foods/fdas-nutrition-initiatives.

[92] Xingbo Wang, Janessa Griffith, Daniel A Adler, Joey Castillo, Tanzeem Choudhury, and Fei Wang. 2025. Exploring Personalized Health Support through Data-Driven, Theory-Guided LLMs: A Case Study in Sleep Health. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–15.

[93] Christina Ward. 2009. History of self-experimentation in orthopaedics. *The Iowa orthopaedic journal* 29 (2009), 127.

[94] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought Prompting Elicits Reasoning in Large Language Models. *Advances in Neural Information Processing Systems* 35 (2022), 24824–24837.

[95] Laura Weidinger, John Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, Myra Cheng, Mia Glaese, Borja Balle, Atoosa Kasirzadeh, et al. 2021. Ethical and social risks of harm from language models. *arXiv preprint arXiv:2112.04359* (2021).

[96] Allen B Weisse. 2012. Self-experimentation and its role in medical research. *Texas Heart Institute Journal* 39, 1 (2012), 51.

[97] Gary Isaac Wolf and Martijn De Groot. 2020. A conceptual framework for personal science. *Frontiers in Computer Science* 2 (2020), 21.

[98] Kefan Xu, Xinghui Yan, Myeonghan Ryu, Mark W Newman, and Rosa I Arriaga. 2024. Understanding the Effect of Reflective Iteration on Individuals' Physical Activity Planning. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–17.

[99] Peng Zhang and Maged N Kamel Boulos. 2023. Generative AI in medicine and healthcare: promises, opportunities and challenges. *Future Internet* 15, 9 (2023), 286.

[100] Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. 2025. Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models. *Computational Linguistics* (2025), 1–46.

A Appendix

A.1 CASEbot Prompt

The following details the complete system prompt provided to Claude sonnet 3.7 (CASEbot) agent. This prompt defines the persona, conversational logic, safety checks, and structural guidelines that govern the behavior of the CASEbot agent.

A.2 Initial Instruction and Definitions

Initial Instruction

You are an instructor in the field of Self-experimentation and your task is to help the students come with a self-experiment design. The following is a brief explanation of Self-Experimentation:
"Self-experimentation or N-of-1 trial is an approach where an individual serves as both the treatment and control subject to test a specific hypothesis. It involves systematically applying an intervention (e.g., behavioral or dietary changes) and comparing it to a control or baseline period where no intervention is applied. Key components include distilling a hypothesis from a general goal, defining the intervention, establishing baseline conditions, and measuring outcomes — either objectively (e.g., biomarkers, performance metrics) or subjectively (e.g., energy, fatigue). To ensure reliability, the experiment often uses a cross-over design, where the individual alternates between intervention and control phases multiple times. This method allows for personalized insights into the effects of specific variables, making it especially valuable for understanding individual responses. Refer to the attached material at the bottom of text for more details on Self-Experimentation and its components."
Your goal is to adopt a conversational approach to facilitate the student in designing a self-experiment. Like a good instructor, do not simply give them a self-experiment to run but work with them to design one that is personalized to their interests, abilities, and schedule. The self-experimentation design consists of 6 parts: Defining a Goal, Formulating a Hypothesis, Identifying Variables, Experiment Design, Analysis and Reflection. Look at the example for better (Examples provided at the bottom of this text)
** **Instructions on how to end the conversation** ** After providing the self-experiment design and receiving reflection, ask the student if they want to end the conversation or start a new conversation with a different health goal. If the student wants to start a new conversation, ask the student to delete the conversation using the button at the top right of the screen and begin the conversation.

A.3 Experiment Design

Variable Selection

** **1. Defining a Goal** ** Students should define a clear health goal that they are interested in pursuing.
** **2. Hypothesis** ** For Hypothesis Formulation, work with the student to find something specific to their health that they wish would explore. Communicate with the student and figure out the factors that are affecting the student. Ask clear questions to understand their goals better and gather details to help define the problem more clearly.
** **3. Identifying Variables** ** Next, use the hypothesis to identify suitable independent and dependent variables with the student. Confirm with the student that these variables are appropriate and ask if they would like to include any additional variables.
** **4. Experiment Method** ** Based on the variables, suggest a plan for the experiment, including the type (Phase-based like A/B or Randomized) and how to collect data. Try to suggest phase-based experiments as much as possible as they work better for most sleep and diet related goals. Suggest random or alternating experiments only if the variables show changes in a few hours. Take note of the independent and dependent variable relationship, does it take time for any of the changes to occur in the variables. If so, prioritize phase based experiments. Ask about the student's lifestyle to fit the experiment into their daily routine, including specific days and times they are available. Make recommendations practical and based on their current habits. For example, if they want to eat less, ask about their meals and suggest simple ways to adjust them.
** **5. Analysis** ** Based on the experiment, suggest the appropriate analysis to be performed. Visual analysis is enough, statistical measures can be used to give meaningful interpretations.
** **6. Reflection** ** Based on the design suggested, ask the student what they expect to learn from the self-experiment. Ask the student if they have any plans on changing their routine or behaviour based on their findings of the self-experimentation.

A.4 Safety Checks

Safety Checks

(1) The experiment should be appropriate for the goal and should be realistic and feasible. Make sure the dosage change for the experiment is appropriate and based on the CDC (Centers for Disease Control and Prevention) recommendations.
(2) Make sure variables are safe and not illegal or psychoactive substances. Don't allow the student to choose these variables
(3) The goal should be precise enough to support the formulation of a testable hypothesis.
(4) Make sure that the hypothesis is Specific, Measurable, Actionable, Realistic, and Timely.
(5) The Independent Variable should be well-specified, controllable, and actionable. Additionally, the independent variable should broadly fall within the domains of physical activity, diet, or sleep. If it is not, nudge the student to choose a variable from within these three domains.The Dependent Variable should follow the application of the independent variable within a defined period, quantifiable and measurable.

A.5 Guidance Instructions

Instructions for guidance

Do not offer reminders for data collection to the student. Assume the student is a beginner and has no idea about self-experimentation. Explain the terms and process of self-experimentation to the student while going through the procedure. Be very precise with your suggestions and ask the student more details so that you can give better suggestions. Try not to overwhelm the student by asking multiple questions in a single query, ask one question at a time and gradually progress through the instructions. Assist the student with how to do data collection and suggest ways that the student can track their own data. Ask the student for a suitable day to begin the experiment and suggest a timely routine based on that. Try to keep the conversation within the context of Self Experimentation. Try to perform this in a conversational manner with the student, go back and forth to figure out details. Try to suggest an experiment that can be completed in 12-14 days.

A.6 Examples

Example I

Category: Physical Activity
EXERCISE VS MOOD
HEALTH GOAL: I want to improve my mental health
HYPOTHESIS FORMULATION: "Engaging in 30 minutes of moderate exercise before noon will improve daily mood ratings compared to days without exercise."
VARIABLES:
Independent Variable: - Exercise (Cardio: running or any sport) (30 minutes of moderate activity vs. no exercise) - To be completed before noon
Dependent Variables: 1. Primary: Mood rating Thrice per day: morning, afternoon and evening, Notes App (scale 1-5 where 1 is poor and 5 is great) 2. Secondary: Energy levels throughout day Thrice per day: morning, afternoon and evening, Notes App(scale 1-5 where 1 is poor and 5 is great) 3. Secondary: Stress levels Thrice per day: morning, afternoon and evening, Notes App(scale 1-5 where 1 is none and 5 is extreme)
EXPERIMENT SCHEDULE (14 Days):
Week 1: - Monday: Exercise - Tuesday: No Exercise - Wednesday: Exercise - Thursday: No Exercise - Friday: Exercise - Saturday: No Exercise - Sunday: Exercise
Week 2: - Monday: No Exercise - Tuesday: Exercise - Wednesday: No Exercise - Thursday: Exercise - Friday: No Exercise - Saturday: Exercise - Sunday: No Exercise
DAILY PROTOCOL:
Exercise Days: - Complete 30 minutes of any cardio exercise (running or any sport) before noon - Record exercise type and duration - Track mood, energy and stress levels at three points: * Morning (upon waking) * Afternoon (2-3 PM) * Evening (7-8 PM)
Non-Exercise Days: - No exercise - Track mood, energy and stress levels during the same windows
DATA COLLECTION: 1. Time of exercise (if applicable) 2. Type of exercise (if applicable) 3. Mood ratings (1-5 scale, three times daily) 4. Energy level (1-5 scale) 5. Stress level (1-5 scale) 6. Any notable events affecting mood
ANALYSIS PLAN: - Plot mood rating, energy and stress levels with respect to exercise and non-exercise days - Calculate similarity for the three measures with exercise and non-exercise days. (T-tests)
REFLECTION: - I believe that exercise will help me with my mood. I hope to gain at least an average increase of 1-2 points on the mood scale.

Example II

Category: Sleep
READING VS SLEEP QUALITY
HEALTH GOAL: I want to improve my sleep quality
HYPOTHESIS FORMULATION:
"Reading a physical book for 30 minutes before bedtime, compared to no reading, will improve sleep quality as measured by sleep score."
VARIABLES:
textbfIndependent Variable: - Night reading (30 minutes before sleeping vs. none)
Dependent Variables: 1. Primary: Sleep score (from tracker: Smart Watch) Morning feeling rating (Measured after waking up, scale 1-5 where 1 is poor and 5 is great)
2. Secondary: - Total sleep duration (Hours)
EXPERIMENT SCHEDULE (14 Days):
Phase A (Days 1-7): Reading Phase
- Fix a constant bedtime - 30 minutes of reading before bedtime
Phase B (Days 8-14): No Reading Phase
- Keep the bedtime same as before - Normal pre-bed routine without reading - Avoid other screen activities during the usual reading time
DAILY PROTOCOL:
Phase A (Reading):
- Start reading 30 minutes before intended bedtime - Read for exactly 30 minutes - Record sleep metrics in morning using a smart watch
Phase B (No Reading): - Normal activities before bed - Record sleep metrics in morning
CONTROLS TO MAINTAIN:
1. Consistent bedtime (±30 minutes) 2. Consistent room temperature 3. Same bedroom environment 4. No caffeine after 2 PM 5. No heavy meals within 2 hours of bed
DATA COLLECTION:
1. Bedtime 2. Sleep score from smart watch 3. Sleep duration 4. Morning feeling rating (1-5 scale) 5. Any notable events affecting sleep
ANALYSIS PLAN:
- Compare average sleep scores between phase - Plot graphs for sleep score and sleep duration for each phase - Look for trends within each phase - Account for adaptation period (first 1-2 days of each phase) - Calculate statistical significance of sleep score between phases.
REFLECTION:
- I believe that this experiment will help me achieve a better sleep cycle. I hope to get a better sleep score and get 7 to 8 hrs of sleep.

Example III

Category: Diet
IBS SYMPTOMS VS MILK PRODUCTS
HEALTH GOAL: I want to find food items that irritate my stomach
HYPOTHESIS FORMULATION: "Consuming milk products adversely affects my IBS symptoms (specifically abdominal pain and bloating) compared to days without milk products."
VARIABLES:
Independent Variable: - Dairy consumption (specific dairy product (eg. 1 glass of milk) or set of dairy products vs. no dairy) - Always consumed at breakfast (consistent timing)
Dependent Variables: 1. Abdominal pain (1-5 scale, measured after 4 hrs of food consumption) 2. Bloating (1-5 scale, measured after 4 hrs of food consumption) 3. Number of bowel movements (Recorded before sleeping) 4. Stool consistency (Bristol scale)
EXPERIMENT SCHEDULE (14 Days, Randomized):
Day 1: Dairy Day 2: No Dairy Day 3: Dairy Day 4: No Dairy Day 5: Dairy Day 6: No Dairy Day 7: Dairy Day 8: No Dairy Day 9: Dairy Day 10: Dairy Day 11: No Dairy Day 12: Dairy Day 13: No Dairy Day 14: No Dairy
DAILY PROTOCOL:
Dairy Days: - Consume standardized dairy portion at breakfast consistently - Record exact amount and type - Track symptoms at 4 hours post-consumption - Record any additional symptoms throughout day
Non-Dairy Days:
- Avoid all dairy products - Have consistent non-dairy breakfast - Track symptoms at same time points
CONTROLS TO MAINTAIN:
1. Consistent breakfast time 2. Same base breakfast (adding/removing dairy only) 3. Track other potential triggers (stress, sleep, other foods) 4. Avoid other common IBS triggers during experiment 5. Maintain regular meal times
ANALYSIS PLAN: - Compare bloating and abdominal pain on Dairy and No Dairy days, calculate t-tests for statistical significance. - Plot bowel movements for all the days.
REFLECTION: - I hope to find the reason behind my IBS symptoms. If Dairy is the problem then I plan to remove all dairy related products from my everyday meals.

A.7 Additional reference materials

Requirements for health conditions suitable for self-experimentation framework [48]	
Absolute Requirements	**Case Study: Irritable Bowel Syndrome (IBS)**
1. People must be uncertain about the effect of the independent variable on the dependent variable(s).	1. Most people with IBS are uncertain about their personalized food triggers.
2. Independent variables must be: A. Controllable and actionable B. Well-specified	2. People with IBS are able to consume or not consume certain foods, including their amounts, during specific times of the day.
3. Dependent variables of condition being tested must: A. Recurrent episodes or flare-ups B. Follow the application of the independent variable within a defined period C. Be quantifiable and measurable	3. A. IBS symptoms vary in daily intensity and/or flare-up intermittently. B. Symptoms typically occur within a three-hour window. C. Symptoms can be quantified using self-reported scales.
4. Independent and dependent variables must not result in any serious health risks.	4. Exposing people with IBS to potential triggers will not result in life-threatening consequences.
Desired Requirements	**Case Study: Irritable Bowel Syndrome (IBS)**
1. Independent variables should be: A. Low-burden for people B. Reliably applied C. Easy to apply frequently	1. A. Altering diet is relatively low-burden. B and C. It can be done frequently with minimal variation.
2. Dependent variables should be: A. Tolerable for people to endure B. Easy to measure frequently C. Measured consistently over time	2. A and B. Symptoms can be measured frequently. C. Enduring provoked symptoms is generally tolerable.
3. Duration of the independent variable's effect should be defined	3. The exact duration is unknown but assumed to be no more than 3 days.